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Serotonin Acting via Cyclic AMP Enhances Both the Hyperpolarizing and Depolarizing Phases of Bursting Pacemaker Activity in the *Aplysia* Neuron R15

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Bath application of 5-HT, at concentrations below 10 μM, enhances the amplitude of the interburst hyperpolarization in the *Aplysia* bursting pacemaker neuron R15. It is known that 5-HT acts via cyclic AMP to produce this effect by increasing the inwardly rectifying potassium current (I_R). Here, we report that further elevating the concentration of 5-HT produces an enhancement of the depolarizing phase of the burst cycle that can eventually lead to tonic spiking activity. Voltage-clamp studies reveal that high concentrations of 5-HT continue to increase I_R and, in addition enhance a voltage-gated inward current active near the action potential threshold. Pharmacological treatments and ion substitution experiments demonstrate that the inward current increased by high concentrations of 5-HT is a subthreshold calcium current (I_{Ca}).

The 5-HT-induced increase in I_{Ca} is mimicked by bath application of the adenylate cyclase activator forskolin or injection of 8-bromo-cyclic AMP and is potentiated by the phosphodiesterase inhibitor isobutylmethylxanthine. It is concluded that 5-HT, acting via the second messenger cyclic AMP, can increase both potassium and calcium currents in neuron R15. It is also shown that the 5-HT-induced increase in these 2 opposing voltage-gated currents not only produces complex changes in bursting activity, but also dramatically alters R15's response to inhibitory and excitatory stimuli.

The second messenger cyclic AMP mediates the actions of a wide variety of neurotransmitters and hormones on excitable cells (Siegelbaum and Tsien, 1983; Levitan, 1985). The actions of cyclic AMP vary between cell type and even within the nervous system of a single animal. In *Aplysia,* cyclic AMP produces excitatory effects in bag cell, sensory, RB, LB, and medial neurons, while it can inhibit spontaneous bursting in cell R15 (Kaczmarek et al., 1978; Klein and Kandel, 1978; Drummond et al., 1980; Pellmar, 1981; Kehoe, 1985a, b, 1986). The varying effects of cyclic AMP occur because cyclic AMP can modulate different ion channels in different neurons. Cyclic AMP has been shown to modulate 6 types of potassium channels, as well as calcium, sodium, and chloride channels in molluscan neurons (Aldenhoff et al., 1983; Ewald and Eckert, 1983; Green and Gillette, 1983; Siegelbaum and Tsien, 1983; Connor and Hock berger, 1984; Hockberger and Connor, 1984; Kaczmarek and Strumwasser, 1984; Kehoe, 1985a, b, 1986; Lotshaw et al., 1986; Strong and Kaczmarek, 1986). In some cells, more than one ion channel is modulated. For example, cyclic AMP decreases 3 distinct potassium currents in *Aplysia* bag cell neurons to produce excitation (Kaczmarek and Strumwasser, 1984; Strong and Kaczmarek, 1986). In *Aplysia* medial neurons, 2 potassium currents are decreased and a sodium current increased (Kehoe, 1985a, b, 1986). Finally, in neuron R15, cyclic AMP increases the inwardly rectifying and calcium-activated potassium currents (Benson and Levitan, 1983; Ewald and Eckert, 1983). In all these cases, cyclic AMP produces its effects by modulating several ion channels which tend to act synergistically to produce either depolarization or hyperpolarization.

5-HT, at concentrations below 10 μM, acts via cyclic AMP to enhance the interburst hyperpolarization produced by the bursting pacemaker *Aplysia* neuron R15 (Drummond et al., 1980). 5-HT causes this effect by increasing the inwardly (or anomalously) rectifying potassium current (I_R) (Benson and Levitan, 1983). Here, we report that higher concentrations of 5-HT also enhance the depolarizing phase of bursting pacemaker activity and can eventually lead to tonic spiking activity. This shift from bursting to beating activity is accompanied by an increase in the subthreshold calcium current (I_{Ca}). The 5-HT-induced increase in calcium current is shown to be mediated by cyclic AMP. These results demonstrate that 5-HT, acting via a single second messenger, cyclic AMP, can produce complex changes in bursting activity by increasing voltage-gated calcium and potassium currents. It is also shown that this modulation of opposing conductances alters the cell's responsiveness to inhibitory and excitatory stimuli.

Materials and Methods

Aplysia californica weighing between 100 and 500 gm were obtained from Alacrity Marine Biological Services (Redondo Beach, CA) and maintained at 16°C. Animals were injected with 50% of their body weight of 400 mM $MgCl_2$. The abdominal ganglion was then removed and the connective tissue sheath overlying cell R15 dissected away. In preliminary experiments, dissection of the sheath was facilitated by treating the ganglion for 1–10 min with 0.25% pronase (Calbiochem). All experiments were reproduced without enzyme treatment.

Cell R15 was identified by its location, pigmentation, and electrical activity (Frazier et al., 1967). R15 was impaled with 1–5 MΩ electrodes containing either 1.5 or 3 M KCl for measuring voltage and similar electrodes containing KCl or 0.5 M K_2SO_4 for passing current. Iontophoretic injection was performed under voltage clamp (Almost Perfect

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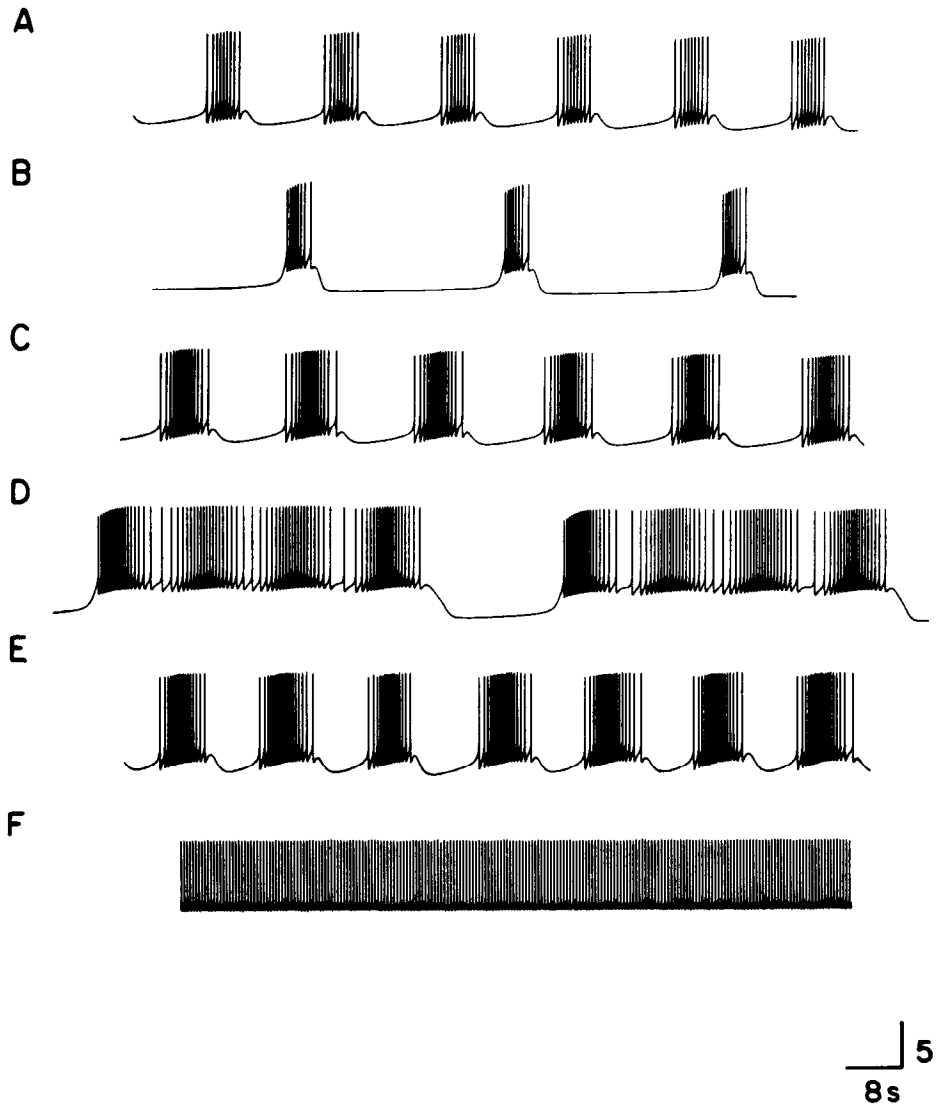


Figure 1. Effects of different concentrations of 5-HT on the bursting activity of cell R15. *A*, Control activity in 100 mM Mg SW. Peak interburst hyperpolarization was −46 mV. *B*, Same as *A* after 12 min exposure to 10 μM 5-HT. Peak interburst hyperpolarization was −68 mV. *C*, After washout of 5-HT. *D*, After 12 min exposure to 50 μM 5-HT. Peak interburst hyperpolarization was −64 mV. *E*, Control activity in another cell in ASW. *F*, Same as *E* after 15 min exposure to 25 μM 5-HT.

Electronics) from a third electrode containing 0.75 M potassium EGTA, pH 8, 2 M Cs_2SO_4, or 0.1 M sodium 8-bromo-cyclic AMP. Iontophoresis of cesium and EGTA simultaneously was accomplished by using a Cs_2SO_4 current-passing electrode and an EGTA-filled iontophoresis electrode. Effectiveness of EGTA injections (typically 500 nA for 5 min) was verified by testing whether calcium-dependent tail currents and hyperpolarizations were blocked. Cesium was iontophoresed at 500 nA for 14 min.

The ganglion was superfused constantly at 1 ml/min in a 1 ml bath with artificial seawater (ASW) containing 460 mM NaCl, 10 mM KCl, 11 mM $CaCl_2$, 55 mM $MgCl_2$, 1% glucose, and 10 mM HEPES, pH 7.4. 5-HT was applied for 10 min unless otherwise stated. To decrease spontaneous synaptic activity, the concentration of $MgCl_2$ was raised to 100 mM at the expense of NaCl (100 mM Mg SW). No qualitative differences in the voltage-clamp data were observed in this solution. In other voltage-clamp experiments where maximal block of synaptic activity was desired without completely blocking subthreshold calcium channels, 2 solutions were used. One was similar to ASW except that the concentration of $MgCl_2$ was raised to 230 mM at the expense of NaCl (230 mM Mg SW). The other solution, which was similar to that used by Mayeri et al. (1985), contained 430 mM NaCl, 10 mM KCl, 75 mM $MgCl_2$, 0.3 mM $CaCl_2$, 3 mM $MnCl_2$, 1% glucose, and 10 mM HEPES, pH 7.4 (0.3 mM Ca SW). The following solution (low Cl SW) was used when a lower external chloride concentration was desired: 392 mM sodium methylsulfate, 10 mM KCl, 50 mM $MgCl_2$, 50 mM $MgSO_4$, 11 mM $CaCl_2$, 1% glucose, and 10 mM HEPES, pH 7.4. Tetraethylammonium bromide (TEA), 4-aminopyridine, manganese chloride, and cadmium chloride were added hypertonically. Cobalt was substituted for calcium. Forskolin was added from a stock 30 mM solution in ethanol. Equivalent exposures of ethanol alone produced no effects. In some experiments, 0.5 mM hexamethonium bromide was included to block cholinergic synapses on R15.

5-HT (5-hydroxytryptamine creatinine sulfate), dopamine (3-hydroxytyramine hydrochloride), sodium 8-bromoadenosine-3′,5′-cyclic phosphate (8-Br-cyclic AMP), 3-isobutyl-1-methylxanthine (IBMX), *N*-2-hydroxyethylpiperazine-*N*′-2-ethanesulfonic acid (HEPES), and bistrispropane were obtained from Sigma. Forskolin was obtained from Calbiochem. Sodium methylsulfate was obtained from Kodak.

All experiments were replicated a minimum of 3 times, and the data shown are representative. The baseline bursting activity and voltage-clamp currents were stable over many hours, and all effects shown were readily reversible upon washing.

Results

5-HT can enhance both the depolarizing and hyperpolarizing phases of R15's bursting pacemaker activity

Bath application of 10 μM 5-HT leads to an enhancement of the depth and duration of the interburst hyperpolarization and can sometimes cause cessation of bursting activity in the *Aplysia* neuron R15 (Drummond et al., 1980). This is due to a cyclic

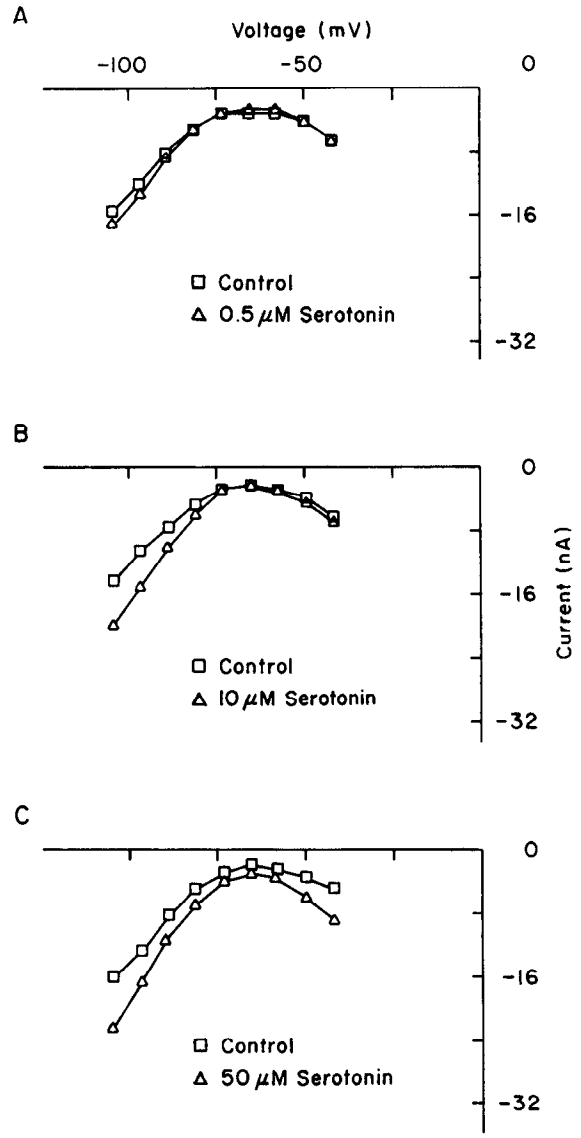


Figure 2. Effects of serotonin (5-HT) on the current versus voltage curve of neuron R15. The effects of 0.5 μM (*A*), 10 μM (*B*), and 50 μM (*C*) 5-HT on current evoked at the end of 1 sec pulses from a holding potential of −50 mV (□, control; △, 5-HT). All data from a single cell.

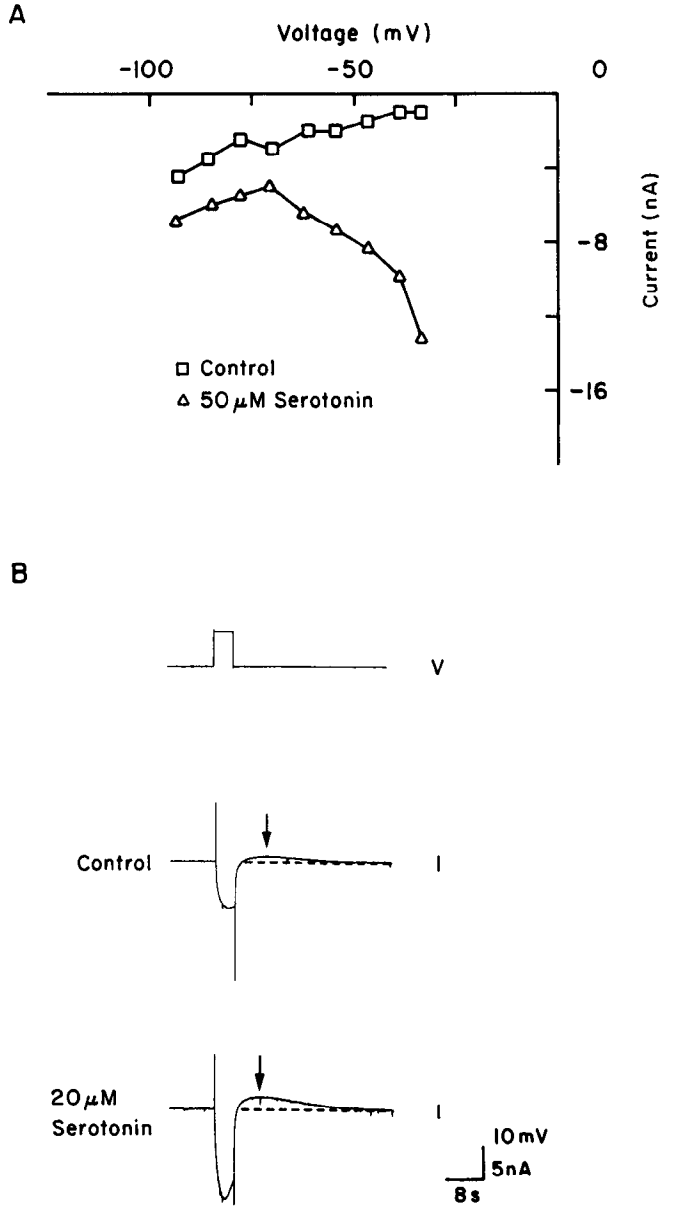


Figure 3. Effect of serotonin (5-HT) on the voltage-gated inward current and the calcium-dependent outward tail current. *A*, I_R was blocked by reducing external potassium to 2 mM in 0.3 mM Ca SW. Currents were evoked with 2 sec pulses from a holding potential of −40 mV: □, control; △, after 50 μM 5-HT. *B*, *Top trace* shows a 10 mV depolarizing voltage pulse from −45 mV; *middle trace*, the evoked current in the control; *bottom trace*, evoked current after 20 min exposure to 20 μM 5-HT. Note that 5-HT increases the voltage-gated inward current, as well as the slow tail current (*arrows*) that is outward relative to the holding current (*dotted line*).

AMP-dependent increase in I_R (Benson and Levitan, 1983). 5-HT at 500 μM, on the other hand, has been found to depolarize R15 under current-clamp conditions (Tremblay et al., 1976). We have found that bath application of much lower concentrations of 5-HT (i.e., 25–50 μM) can produce an enhancement of both the depolarizing and hyperpolarizing phases of the burst cycle. Figure 1*A* shows the basal bursting activity of R15 in the presence of elevated magnesium. Upon bath application of 10 μM 5-HT an enhancement of the interburst hyperpolarization is seen (Fig. 1*B*). After washing out the 5-HT and regaining basal bursting activity (Fig. 1*C*), 50 μM 5-HT was applied. Once again, an enhancement of the interburst is seen, this time accompanied by a large increase in the number of action potentials during the depolarizing phase (Fig. 1*D*). In normal ASW this depolarizing effect predominates so that the cell eventually ceases bursting and instead spikes tonically at a high frequency (Fig. 1*F*). Thus, increasing the concentration of 5-HT can convert its effect from a predominantly inhibitory one to a predominantly excitatory one.

5-HT increases the subthreshold voltage-gated calcium and inwardly rectifying potassium currents

Voltage-clamp experiments were carried out to determine the ionic mechanism by which 5-HT modulates the bursting activity of R15. Bath application of 0.5 μM 5-HT produces an increase in a potassium current that activates with hyperpolarization [the inwardly rectifying potassium current (I_R); Benson and Levitan,

1983]. This change in current is most apparent at potentials more hyperpolarized than −75 mV, the potassium equilibrium potential (Fig. 2*A*). It is thought that increasing I_R leads to enhancement of the interburst hyperpolarization. Raising the concentration of 5-HT to 2–10 μM further increases I_R (Fig. 2*B*). In addition, this effect is accompanied by a small increase in the inward current at potentials more positive than −65 mV, the negative slope resistance region of the current versus voltage (*I-V*) curve (Fig. 2*B*). Increasing 5-HT to 50 μM produces little further change in I_R, indicating that the excitatory effects of 5-HT are not due to a reversal of the increase in that current. However, an additional increase in inward current is seen at depolarized potentials (Fig. 2*C*). This effect is also seen under conditions that dramatically lower synaptic activity but that leave calcium currents partially intact (Fig. 3*A*), as well as in totally isolated neuron R15 in culture (Lotshaw et al., 1986). Hence, it results from a direct effect of 5-HT on R15.

Blocking I_R with either low potassium (Fig. 3*A*) or barium (data not shown) reveals that 5-HT apparently increases 2 additional currents. One component is clearly voltage-gated and displays slow kinetics (Fig. 3*B*). It begins to activate between −70 and −60 mV and increases with depolarization. These characteristics are reminiscent of the subthreshold calcium current. A voltage-independent current is seen at more hyperpolarized potentials (Fig. 3*A*) and may be due to activation of a voltage-independent sodium current (Connor and Hockberger, 1984) or to activation of the same voltage-gated current described above but at sites remote from the cell body where voltage cannot be controlled. In favor of the latter interpretation is the finding that the voltage-independent current is not seen when R15 is maintained in primary cell culture and processes are removed prior to voltage clamping (D. P. Lotshaw and I. B. Levitan, unpublished observations). Furthermore, agents that reduce the voltage-gated current, such as cobalt, manganese, or dopamine (see below), also reduce the voltage-independent current induced more negative than −70 mV. Thus, it seems likely that 5-HT induces only one type of current in addition to I_R. We concentrated on studying the voltage-gated conductance at depolarized potentials which probably is responsible for producing 5-HT's excitatory effect.

The subthreshold inward current active at potentials more positive than −75 mV has been shown to be due to voltage-gated calcium channels that are inactivated by elevated internal calcium (Gorman et al., 1982; Lewis et al., 1984; Adams and Levitan, 1985; Kramer and Zucker, 1985b). Action potentials or depolarizing voltage pulses that allow calcium influx produce a slow outward tail current, which has been shown to be due to calcium-dependent inactivation of resting calcium current normally active at subthreshold voltages (Adams and Levitan, 1985; Kramer and Zucker, 1985b). We found that this outward tail current is increased along with the voltage-gated inward current by 5-HT (arrows in Fig. 3*B*). This effect is also seen in high-magnesium seawater (data not shown), where synaptic input is minimized (Mayeri et al., 1985). This result is consistent with the hypothesis that 5-HT increases a calcium current (I_{Ca}).

In order to test further whether 5-HT increases subthreshold I_{Ca}, calcium currents were blocked with either manganese, cobalt, or cadmium. Shown in Figure 4*A* is the effect of 5-HT on the slow inward current elicited by depolarization in normal saline in a cesium-loaded cell. As in Figure 3*B*, 5-HT nearly doubles the voltage-gated inward current. After washing out the 5-HT, manganese was added. This blocks the voltage-gated inward current (Fig. 4*B*, left). Application of 5-HT in the presence of manganese fails to produce any change in current (Fig. 4*B*, right). Similar results were obtained when calcium was replaced with manganese or cobalt in cells not loaded with cesium (data not shown). Addition of 1 mM cadmium was also found to block the 5-HT induced current (data not shown). These results suggest that 5-HT acts upon I_{Ca} and/or calcium-activated currents.

5-HT might increase the voltage-gated inward current by de-

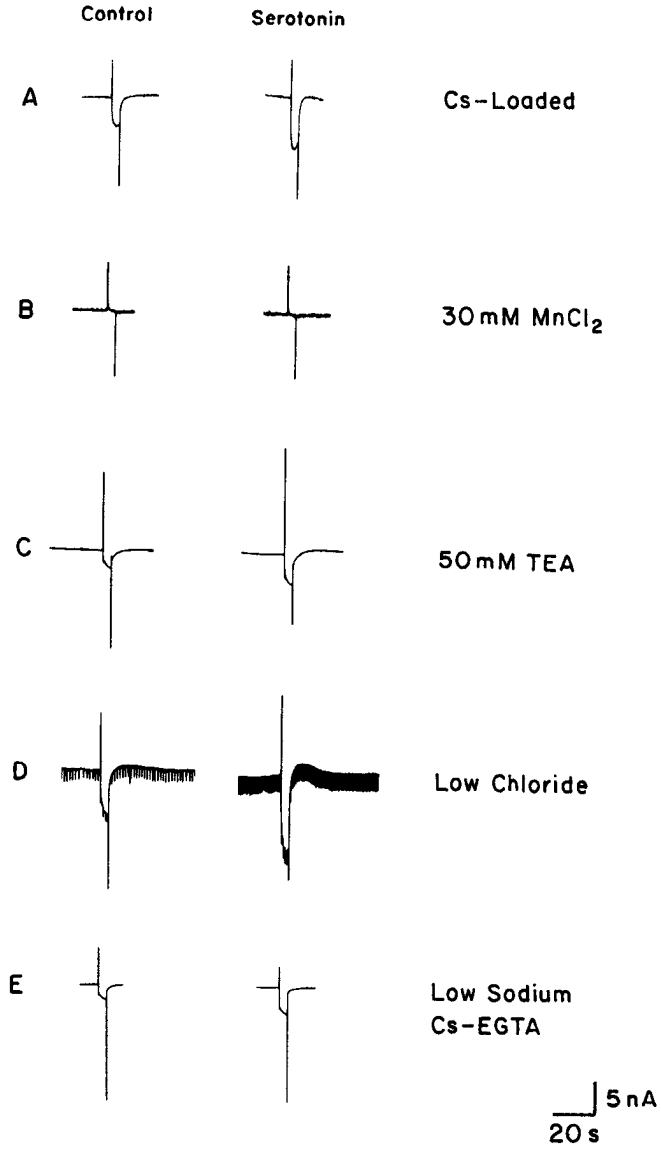


Figure 4. Pharmacology of the serotonin (5-HT)-induced voltage-gated inward current. In each case the current trace at *left* is the control; at *right*, after 3–12 min exposure to 50 μM 5-HT. *A*, Currents evoked by a 10 mV depolarizing pulse from −50 mV in a cesium-loaded cell. *B*, Currents in the same cell in the presence of 30 mM added $MnCl_2$. Similar results were obtained when calcium was *replaced* with manganese or cobalt. *C*, Currents evoked in another cell by a 5 mV depolarizing pulse from −24 mV in 230 mM Mg SW containing 50 mM TEA. *D*, Currents evoked in another cell by a 5 mV depolarizing pulse from −50 mV in low-chloride SW. Downward rapid current deflections are due to a spontaneous cholinergic synaptic input. *E*, After iontophoretically loading R15 with cesium and EGTA, sodium was lowered to 50 mM by substitution with bistrispropane. Currents were then evoked by a 7 mV depolarizing pulse from −36 mV before and after exposure to 5-HT.

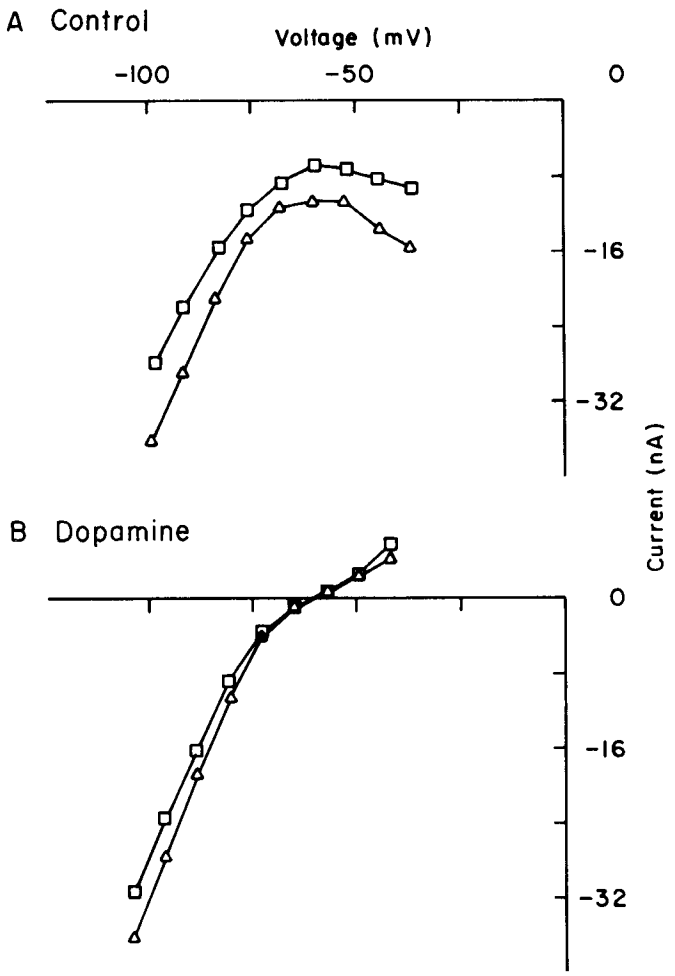


Figure 5. Dopamine attenuates the 5-HT-induced increase in the subthreshold inward current. *A*, Effect of 20 μM 5-HT on R15's *I-V* curve; □, control; △, 5-HT. *B*, Effect of 5-HT in the presence of 250 μM dopamine: □, control; △, 5-HT.

creasing outward potassium currents, although this would produce a decrease rather than the observed enhancement (Fig. 3*B*) of the outward tail current. External application of 50 mM TEA does not block the 5-HT-induced increase in the voltage-gated inward current (Fig. 4*C*) despite the fact that this concentration is sufficient to block 99% of the calcium-activated potassium current and 89% of the delayed rectifying potassium current in R15 (Hermann and Gorman, 1981). Application of 5 mM 4-AP also leaves the 5-HT effect intact (data not shown).

At least 2 potassium currents that are insensitive to TEA and 4-AP but that are blocked by substitution of external calcium with cobalt are known to exist in *Aplysia* neurons. One of these is a TEA-insensitive, calcium-activated potassium channel. Dietmer and Eckert (1985) reported that iontophoretic loading of cells with cesium blocks this current. Likewise, iontophoretically loading tail sensory neurons with cesium blocks a 5-HT-modulated potassium current that is sensitive to cobalt (Pollock et al., 1985). Cesium loading R15 to a similar extent, as judged by the measured block of the delayed rectifying potassium current (data not shown), does not block the 5-HT-induced increase in the subthreshold inward current (Fig. 4*A*). These results lead us to conclude that 5-HT does not act by decreasing potassium currents.

5-HT might produce an apparent increase in the subthreshold inward current by decreasing an opposing outward chloride current. In order to test this possibility, external chloride was reduced from 624 to 132 mM by substituting with methylsulfate, an ion that does not permeate acetylcholine-gated chloride ion channels in *Aplysia* (Kehoe, 1972). This reduction would be expected to shift the chloride equilibrium potential from approximately −58 mV (Russell and Brown, 1972; Ascher et al., 1976) to about −19 mV, a potential well above the spike threshold. 5-HT still produces a robust increase in the inward current at −45 mV despite reversal of chloride currents (Fig. 4*D*), and thus cannot be acting by decreasing a chloride current.

Yet another possibility is that 5-HT activates a voltage-gated sodium current. Reduction of sodium 9-fold does indeed block 5-HT's effect on the inward current at potentials more positive than −75 mV (data not shown). However, it has been shown that removal of sodium also blocks the subthreshold calcium current unless the cell is loaded with EGTA and potassium currents are blocked (Gorman et al., 1982). This may occur because sodium removal might allow calcium entry (perhaps by inhibiting sodium–calcium exchange), which in turn leads to inactivation of I_{Ca} by internal calcium. We have confirmed that iontophoretic loading of R15 with cesium and EGTA, which buffers calcium and blocks potassium channels, dramatically slows the loss of the subthreshold inward current upon reduction of external sodium. Furthermore, under these conditions 5-HT still produces an increase in the subthreshold inward current in low sodium (Fig. 4*E*). This, together with the calcium channel blocker experiments, suggests that 5-HT does not activate a sodium current. Furthermore, 5-HT cannot be modulating some intracellular calcium-dependent process since these are blocked following EGTA loading while the 5-HT effect is still intact (Fig. 4*E*). These results leave the hypothesis that 5-HT increases I_{Ca} as the most viable one. A similar conclusion has been reached from studies of cell R15 maintained in primary culture where pharmacological isolation of calcium currents can be done more rigorously (D. P. Lotshaw and I. B. Levitan, unpublished observations). There it has been shown that 5-HT increases the subthreshold and suprathreshold I_{Ca} without changing its voltage dependence.

This conclusion is further reinforced by the finding that dopamine, which is known to decrease axonal I_{Ca} (Lewis et al., 1984), dramatically reduces the 5-HT-induced increase in inward current. Shown in Figure 5*A* are current versus voltage curves from a cell before and after exposure to 5-HT. The 5-HT-induced increases in I_R and I_{Ca} are evident. Application of a saturating concentration of dopamine blocks the axonal subthreshold I_{Ca}. This is seen in Figure 5*B* as a loss of the inward current at potentials positive to −65 mV. Application of 5-HT in the continued presence of dopamine still produces some small increase in I_{Ca} (above −50 mV) (Fig. 5*B*), but this effect is much less than in the absence of dopamine (Fig. 5*A*). The 5-HT-induced increase in slope conductance below −75 mV, which is due to I_R, remains intact in the presence of dopamine. Thus, a transmitter known to decrease I_{Ca} in R15 reduces the ability of 5-HT to increase the subthreshold inward current. These data not only support the conclusion that 5-HT modulates calcium channels but also suggest that some of the calcium channels activated by 5-HT are inhibited by dopamine. These channels presumably reside in the axon or axon hillock regions of R15 where dopamine is known to act (Ascher, 1972; Lewis et al., 1984).

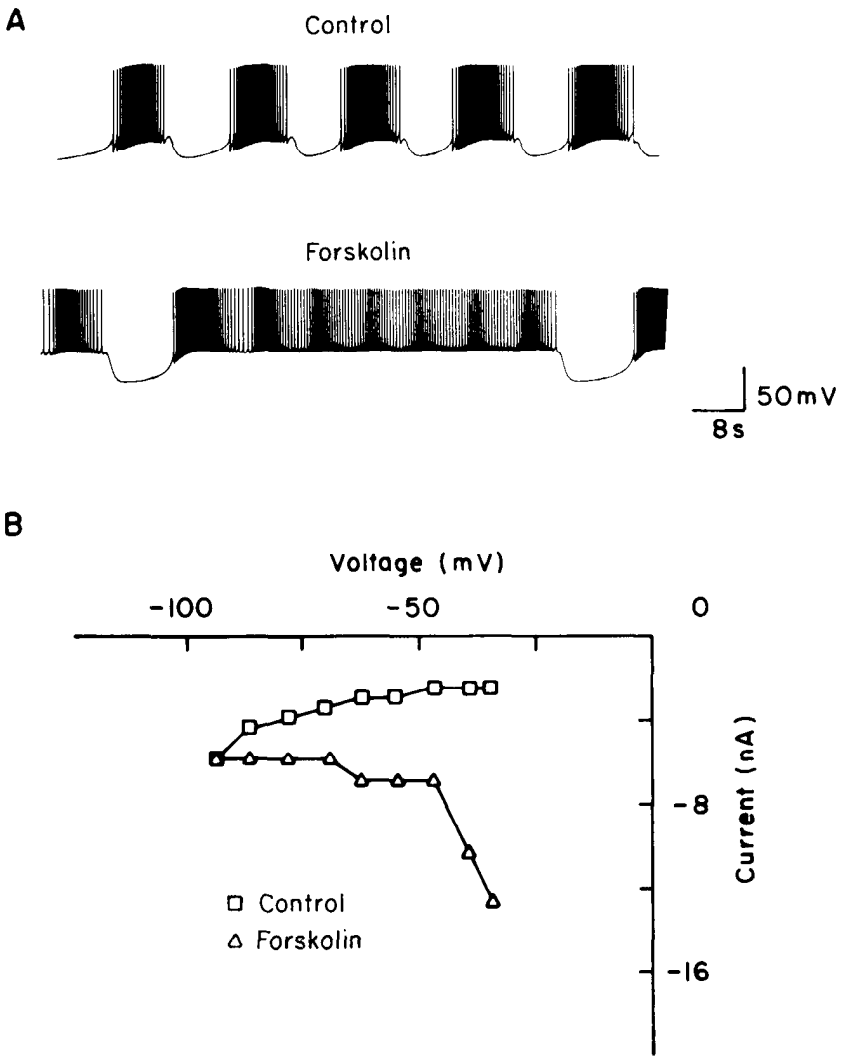


Figure 6. Effect of forskolin on bursting activity and on I_{Ca}. *A, Top trace* shows control bursting activity; *bottom trace,* bursting activity after 30 min in 80 μM forskolin. *B,* Effect of forskolin on the *I-V* curve in 0.3 mM Ca SW containing 2 mM potassium to reduce I_R: □, control; △, after 10 min exposure to 90 μM forskolin.

5-HT acts via cyclic AMP to increase calcium current in R15

It has been demonstrated that 5-HT activates adenylate cyclase and elevates cyclic AMP in R15 (Cedar and Schwartz, 1972; Levitan, 1978). This, in turn, mediates 5-HT's effects on I_R (Drummond et al., 1980; Lemos and Levitan, 1984) via activation of cyclic AMP-dependent protein kinase (Adams and Levitan, 1982). A number of experiments were performed to test whether activation of adenylate cyclase is also responsible for the increase in I_{Ca} seen with higher levels of 5-HT. Forskolin, an activator of adenylate cyclase in many tissues, including *Aplysia* neurons (Seamon et al., 1981; Kauer and Kaczmarek, 1985), was bath applied to R15. Enhancement of bursting activity (Fig. 6*A*) and increases in I_R (data not shown) and subthreshold I_{Ca} (Fig. 6*B*) were seen even under conditions that minimized synaptic activity. Furthermore, the effects of 5-HT and forskolin occlude each other, suggesting that activating adenylate cyclase modulates the same ion channels as 5-HT (data not shown). The possibility that cyclic AMP could mediate 5-HT's effects on R15 was further tested by injecting 8-Br-cyclic AMP into R15. Once again, I_R and I_{Ca} were increased (Fig. 7). These results lead us to conclude that activation of adenylate cyclase and elevation of cyclic AMP can increase subthreshold I_{Ca} in cell R15.

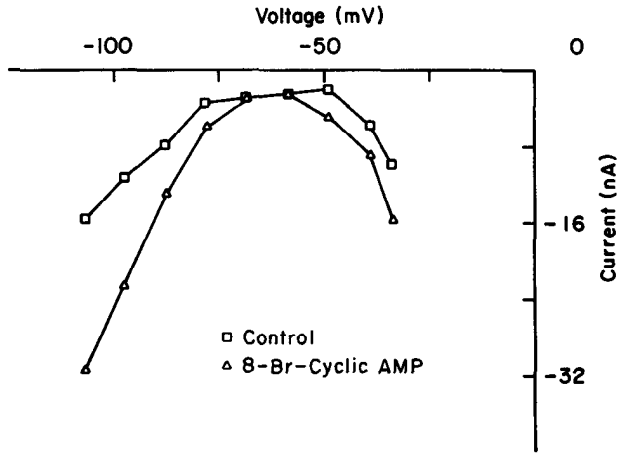


Figure 7. Effect of 8-Br-cyclic AMP injection on the *I-V* curve of R15. Currents evoked by 2 sec pulses from −40 mV before (□) and 4 min after (△) pressure injection of 8-Br-cyclic AMP. The injection electrode contained 10 mM 8-Br-cyclic AMP. Similar results were obtained with iontophoretic injections.

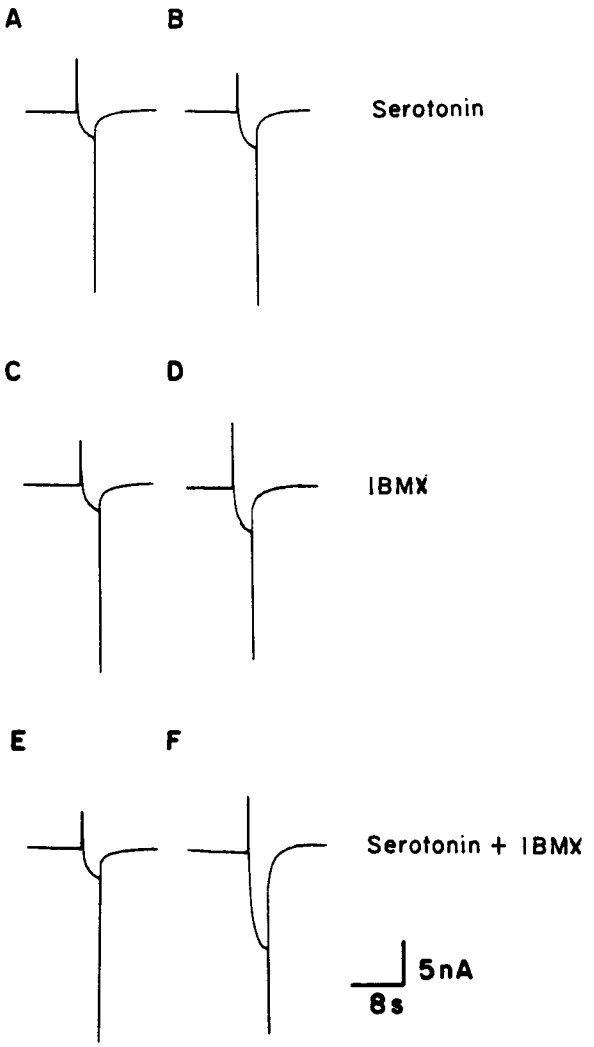


Figure 8. Phosphodiesterase inhibitor potentiates the serotonin (5-HT)-induced increase in I_{Ca}. In each case the current evoked by an 8 mV depolarizing voltage pulse from −48 mV is shown. *A*, Control. *B*, After addition of 1 μM 5-HT. *C*, After washing out the 5-HT. *D*, After addition of 0.2 mM IBMX. *E*, After washing out the IBMX. *F*, After exposure to 1 μM 5-HT together with 0.2 mM IBMX. Note that the increase in current produced by the combined application of 5-HT and IBMX is much larger than the sum of the increases produced by 5-HT and IBMX separately.

It remains possible, however, that 5-HT increases I_{Ca} by some mechanism independent of cyclic AMP. To address this possibility, we conducted experiments with the phosphodiesterase inhibitor IBMX. Figure 8, *A* and *B*, shows the effect of 1 μM 5-HT on I_{Ca}; Figure 8, *C* and *D*, shows the effect of bath application of 0.2 mM IBMX. In both cases, only very small increases in I_{Ca} occurred. Application of 5-HT and IBMX together, however, produced a large increase (Fig. 8, *E*, *F*). Thus, a phosphodiesterase inhibitor potentiates the 5-HT-induced increase in I_{Ca}. This result, taken together with the facts that 5-HT elevates cyclic AMP in R15 and that forskolin and 8-Br-cyclic AMP mimic the effects of 5-HT, leads us to conclude that 5-HT acts via a single second messenger, cyclic AMP, to activate at least 2 ion channels, I_R and I_{Ca}. It may be the balance between these 2 conductances, which have opposing effects on cell excitability, that determines whether a given concentration of 5-HT produces profound inhibition, tonic excitation, or enhancement of both the depolarizing and hyperpolarizing phases of the burst cycle.

R15 responses to inhibitory and excitatory input are altered by 5-HT

R15 often does not exhibit robust bursting activity in the intact nervous system; rather, its activity is sometimes found to be dominated by a constant stream of synaptic input (see Adams and Benson, 1985). Therefore, in order to determine the possible physiological importance of modulating 2 voltage-gated conductances with opposing effects on cell excitability, one must determine how 5-HT alters the sensitivity of R15 to synaptic currents. As a preliminary approach to this question, current pulses were applied to R15 in the hope that these would mimic the effects of synaptic currents. While these stimuli are not physiological, passing current directly into R15 insures that any effects seen arise in R15 and not in a presynaptic cell.

Short hyperpolarizing (Fig. 9, *A*, *B*) and depolarizing (Levitan and Levitan, 1988) current pulses produce little effect on the normal bursting activity in R15. Application of 25 μM 5-HT, however, not only produces tonic firing activity but also dramatically alters R15's response to transient currents. A brief hyperpolarizing current pulse in the presence of excitatory concentrations of 5-HT triggers a long-lasting hyperpolarizing plateau potential (Fig. 9*C*). Increasing the magnitude of the current pulse produces almost no change in the depth and duration of the evoked hyperpolarization. These data suggest that the current pulse serves merely to trigger the activation (and/or inactivation) of voltage-gated currents, which, in turn, regeneratively produce the hyperpolarization.

If this is the case, then any outward current, if sufficiently large, may trigger a long hyperpolarizing plateau potential. Depolarizing current pulses that increase spike frequency produce a slow outward current (Levitan and Levitan, 1988), and this outward current can also trigger a long-lasting hyperpolarizing plateau potential in the presence of high concentrations of 5-HT (Fig. 9*E*; the increase in spike frequency is substantial but is not discernible on this time scale).

5-HT can also hyperpolarize and terminate bursting activity in R15 (Drummond et al., 1980), and under these conditions, depolarizing current pulses can trigger a burst of action potentials (Fig. 9*D*). Once again, this is reminiscent of "plateau potential" cells (Russell and Hartline, 1978; Selverston and Moulins, 1985). These results reinforce the conclusion that 5-HT does not produce simple excitation or inhibition, but rather alters R15's bursting activity and sensitivity to hyperpolarizing or depolarizing input. Effectively, 5-HT converts R15 from an autonomous burster into a plateau potential cell. These properties most likely arise because 5-HT increases 2 discrete voltage-gated currents having opposite effects on cell excitability.

Discussion

We report here that the role of cyclic AMP is more complicated than previously suspected in the bursting pacemaker neuron R15. Application of 10 μM 5-HT, a transmitter known to elevate cyclic AMP in R15 (Cedar and Schwartz, 1972), enhances the interburst hyperpolarization of the bursting pacemaker cycle (Drummond et al., 1980). At higher concentrations, 5-HT induces enhancement of both the depolarizing and hyperpolarizing phases of the burst cycle and often produces tonic spiking activity. Voltage-clamp experiments reveal that the increase in I_R underlying the inhibition of bursting at low concentrations

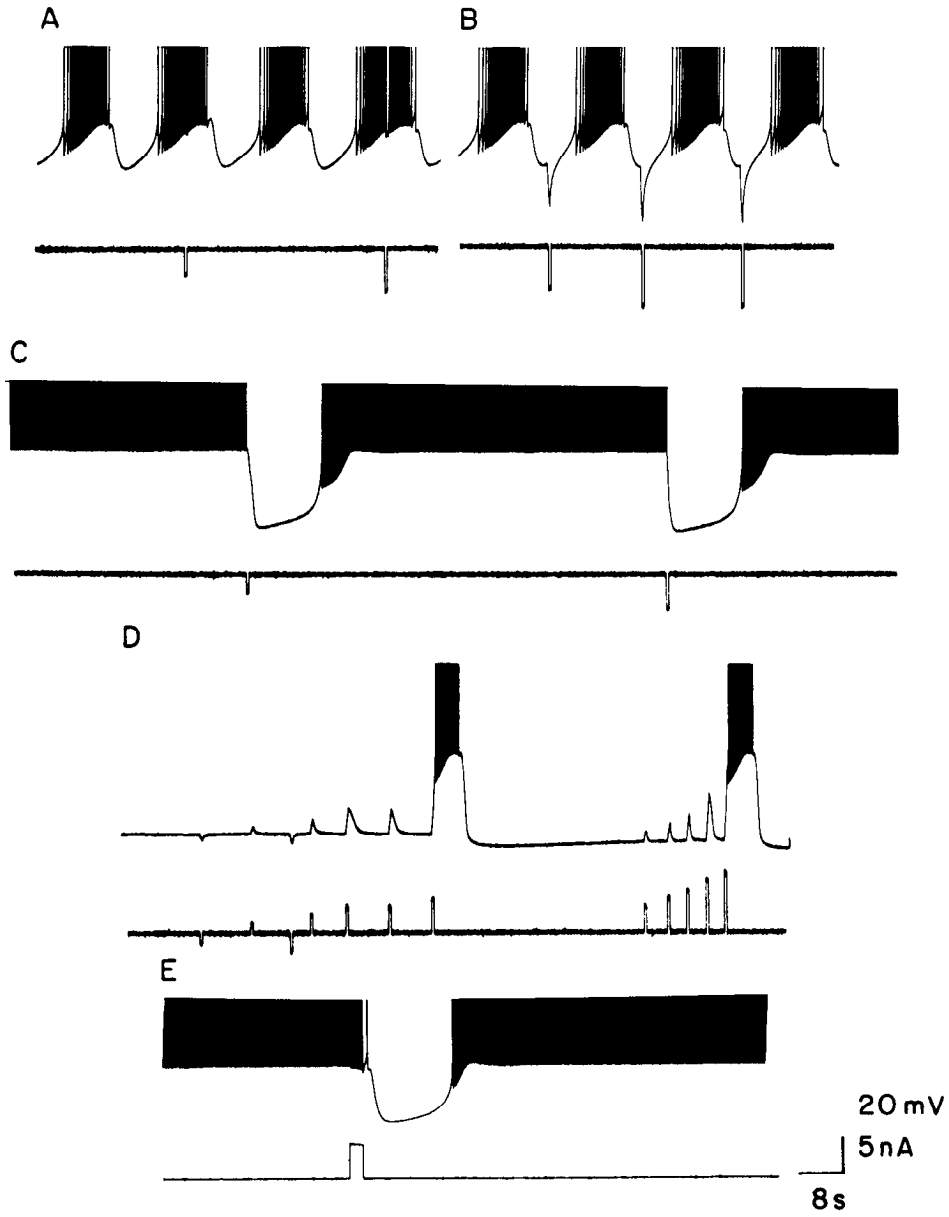


Figure 9. 5-HT alters the effects of hyperpolarizing and depolarizing current pulses on R15's activity. Brief hyperpolarizing current pulses (*lower traces*) applied to a control cell during a burst (*A*) or during the interburst (*B*) produce no long-term changes in R15's bursting activity. *C*, After inducing tonic spiking activity with 25 μM 5-HT, small hyperpolarizing current pulses trigger long-lasting hyperpolarizations. Increasing the size of the pulse has little effect. *D*, While washing out the 5-HT, this cell hyperpolarized, and during this period depolarizing current pulses can trigger a burst. Bursts cannot be triggered in cells hyperpolarized with dopamine. *E*, In another cell, tonic spiking was induced with 30 μM 5-HT. Long-lasting hyperpolarizations could be induced with depolarizing pulses that increase spike frequency. The change in spike frequency is not apparent with the time scale shown. A similar depolarizing pulse has little effect on basal bursting activity (not shown; see Levitan and Levitan, 1988). The tops of the spikes were cut off by the chart recorder in all records.

of 5-HT (Benson and Levitan, 1983) remains intact at higher concentrations. Enhancement of the depolarizing phase of the burst cycle, therefore, is not due to a loss of the cyclic AMP-mediated increase in I_R; rather, 5-HT's depolarizing effect is accompanied by an increase in subthreshold I_{Ca}.

The modulation of I_{Ca} by 5-HT is mimicked by either the adenylate cyclase activator forskolin or 8-Br-cyclic AMP, and it is potentiated by the phosphodiesterase inhibitor IBMX. These data, together with the fact that 5-HT increases cyclic AMP in R15 (Cedar and Schwartz, 1972), lead us to conclude that 5-HT's hyperpolarizing and depolarizing effects are both mediated by cyclic AMP. Thus, 5-HT, acting via a single second messenger (cyclic AMP), can produce complex changes in bursting activity by modulating voltage-gated ionic currents (I_{Ca} and I_R) with opposing effects on excitability.

The suggestion that cyclic nucleotides can be excitatory in R15 is not novel. Bath application of a membrane-permeable cyclic AMP analog can produce excitation in R15. However, its effects were attributed to the fact that it indirectly elevates cyclic GMP by inhibiting phosphodiesterases (Levitan et al., 1979). Likewise IBMX has been reported to cause excitatory effects and an increase in the slow inward current, and it was hypothesized that these excitatory effects were also mediated by cyclic GMP, rather than cyclic AMP (Drake and Triestman, 1981). We have found that a membrane-permeable cyclic GMP analog in fact *decreases* I_{Ca} (Levitan and Levitan, 1988), and thus the excitatory effects of IBMX are probably mediated by cyclic AMP, not by cyclic GMP. This conclusion is reinforced by the fact that R020-1724, an inhibitor specific for cyclic AMP phosphodiesterase, also increases I_R and I_{Ca} (unpublished observations).

We have shown here that 5-HT, acting via cyclic AMP, can increase a potassium and a calcium current, and that these 2 currents are modulated at different concentrations of 5-HT. However, when R15s from juvenile *Aplysia* are maintained in primary cell culture, 1 μM 5-HT is sufficient to produce large increases in both I_R and I_{Ca}. A chloride current, which is evident only when internal chloride is elevated, is also modulated by

5-HT in these cultured cells (Lotshaw et al., 1986). The fact that 1 μM 5-HT activates the different currents equally in cultured cells may be due to a greater sensitivity to 5-HT or, in part, to the fact that cultured cells are completely accessible to 5-HT, whereas *in situ* R15's soma is exposed while the axon is buried in the neuropil. If 5-HT-sensitive potassium channels are preferentially located on the soma, whereas 5-HT-sensitive calcium channels are located on the axon, then the unequal accessibility of the soma and axon *in situ* might produce an apparent difference in sensitivity to 5-HT. Such compartmentation of cyclic AMP-sensitive channels could be tested by locally applying 5-HT to R15's axon and soma.

The question arises of whether the cyclic AMP-induced increases in opposing currents, seen with high concentrations of 5-HT, are physiologically relevant. No known serotonergic neurons synapse on R15, and the levels of circulating 5-HT in *Aplysia* are not well documented. It is known, however, that excitation of the bag cell neurons leads to secretion of a peptide, egg-laying hormone (ELH), that augments both the hyperpolarizing and depolarizing phases of R15's burst cycle (Branton et al., 1978; Mayeri et al., 1985). We have recently found that bag cell discharge or application of pure ELH increases I_R and I_{Ca}, and that these increases are mediated by an increase in intracellular cyclic AMP in R15 (Levitan et al., 1987). Thus, a cyclic AMP-mediated increase in opposing currents represents a physiological situation.

The effects of ELH suggest a possible reason to modulate opposing currents, which in a normally silent cell makes little intuitive sense. I_R activates regeneratively with hyperpolarization at potentials more negative than the action potential threshold (i.e., −35 mV). Therefore, increasing I_R will tend to enhance the interburst hyperpolarization. The subthreshold I_{Ca} activates regeneratively with depolarization near the action potential threshold, and along with a spike-evoked, Ca^{2+}-activated inward current (Lewis et al., 1984; Adams, 1985; Adams and Levitan, 1985; Kramer and Zucker, 1985a), provides the depolarizing drive to maintain a burst. Since I_R and I_{Ca} gate at different voltages, and since the regenerative activation of one inactivates the other, increasing both of these voltage-gated currents tends to enhance both the hyperpolarizing and depolarizing phases of the bursting pacemaker cycle. Because bursting activity is thought to be optimal for producing secretion and synaptic release (Gillary and Kennedy, 1969; Dutton and Dyball, 1979), the enhancement of oscillation produced by increasing opposing currents may lead to an enhancement of secretion by R15, a cell thought to have a neurosecretory function (see Adams and Benson, 1985).

We suggest that another consequence of increasing I_{Ca} and I_R is to amplify R15's sensitivity to synaptic potentials. Normally, small hyperpolarizing current pulses have little effect on bursting activity. However, 5-HT increases I_R, which regeneratively activates with hyperpolarization and drives the cell towards the potassium equilibrium potential (−75 mV). Thus, in the presence of 5-HT, there is a larger amount of I_R to be activated by hyperpolarization [evoked by experimentally injecting current (Fig. 9*C*) or by increasing spike frequency (Fig. 9*E*)], and hence long hyperpolarizations are triggered by these small stimuli. These hyperpolarizations finally end because steady-state inactivation of I_{Ca} is slowly removed at the hyperpolarized potential, causing depolarization; this, in turn, inactivates I_R. These results suggest that the 5-HT-induced increases in 2 voltage-gated currents that have opposite effects on cell excitability may enhance R15's sensitivity to excitatory and inhibitory synapses. In fact, the primary physiological effect of elevating cyclic AMP in R15 may be to convert R15 from a bursting cell to a plateau potential cell (Russell and Hartline, 1978; Selverston and Moulins, 1985).

It is of interest to compare the effects of dopamine and 5-HT. Like 10 μM 5-HT, dopamine enhances the interburst hyperpolarization and can terminate bursting activity (Ascher, 1972). In contrast to 5-HT, dopamine acts by decreasing I_{Ca} (Lewis et al., 1984). Furthermore, R15 cannot be induced to burst with depolarizing current in the presence of high concentrations of dopamine (Wilson, 1980). Hyperpolarizing concentrations of 5-HT, however, do not prevent the triggering of bursts by depolarizing current pulses (Fig. 9*D*). Thus, while 5-HT and dopamine can both inhibit R15's intrinsic bursting activity, they produce opposite effects on R15's responsiveness to excitatory stimuli.

Dopamine can interact with 5-HT by reducing the stimulation of I_{Ca} by 5-HT. This implies that both transmitters can modulate the same calcium channels. It also suggests that 5-HT must elevate cyclic AMP and modulate calcium channels in the axon where dopamine exerts its actions (Ascher, 1972; Lewis et al., 1984). Finally, the fact that dopamine reduces the excitatory effects of 5-HT suggests that small concentrations of dopamine might be sufficient to convert a net excitatory effect of 5-HT and cyclic AMP into a potent inhibitory one. This possibility is of physiological interest since stimulation of the branchial nerve can lead to long-term synaptic inhibition in R15, which appears to consist of a combination of 5-HT-like and dopamine-like effects (Adams et al., 1980).

The actions of cyclic AMP on cell R15 cannot simply be classified as excitatory or inhibitory. This is not wholly surprising since the basal activity of R15 is much more complex than that of some other neurons in which the actions of second messengers have been studied. It will be of interest to determine whether the modulation of opposing conductances by a single second messenger will be found to be a typical, rather than a unique, situation in bursting pacemaker neurons. If so, the fact that bursting pacemaker neurons play a central role in controlling neurosecretion and pattern generation (Rapp, 1979; Carpenter, 1982; Selverston and Moulins, 1985) suggests that this phenomenon may be of wide importance in the operation of the nervous system.

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A Cyclic GMP Analog Decreases the Currents Underlying Bursting Activity in the *Aplysia* Neuron R15

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Bath application of 8-parachlorophenylthio-cyclic GMP (8-pcpt-cGMP) has been shown to increase the number of action potentials per burst in the *Aplysia* neuron R15. Here we report that 8-pcpt-cGMP can eventually inhibit R15's bursting activity and cause the cell to exhibit slow tonic spiking activity. This action is preceded by decreases in spike frequency and in the amplitude of the interburst hyperpolarization. The ionic mechanism underlying these various changes in burst pattern was investigated using voltage clamp. 8-pcpt-cGMP reduces steady-state inward current, and this effect is blocked by manganese but is unaffected by internal EGTA or elevation of extracellular potassium. Dopamine, which decreases the subthreshold calcium current in R15, diminishes the change in current induced by 8-pcpt-cGMP. 8-pcpt-cGMP also reduces the spike-evoked calcium-dependent inward and outward currents known to underlie burst generation. These data lead us to conclude that 8-pcpt-cGMP acts directly to reduce the steady-state subthreshold calcium current (I_{Ca}). This decrease in I_{Ca} may also indirectly reduce the calcium-dependent currents. It is suggested that 8-pcpt-cGMP produces its effects on bursting by decreasing both the inward and outward currents required for generating bursting behavior, and thus effectively dampens bursting until the cell is in a beating state. It is also shown that 8-pcpt-cGMP dramatically alters R15's responses to depolarizing and hyperpolarizing stimuli. Thus, this cyclic GMP analog alters not only the cell's intrinsic activity, but also its response to external inputs.

Bursting electrical activity is produced by many neurons and endocrine tissues (Rapp, 1979; Carpenter, 1982). The importance of bursting in the nervous system is at least 2-fold. First, the rhythmic outputs of bursting cells serve as timers or pacemakers in central pattern generators (Selverston and Moulins, 1985). Bursting activity is also found in many neurosecretory cells presumably because it represents the optimal activity for producing secretion and synaptic release (Gillary and Kennedy, 1969; Dutton and Dyball, 1979). Induction and modulation of bursting activity by synapses and transmitters have been found in both vertebrate and invertebrate nervous systems (Parnas et al., 1974; Wilson and Wachtel, 1978; Grace and Bunney, 1984; Marder and Eisen, 1984; Roth et al., 1984; Dekin et al., 1985; Harris-Warrick and Flamm, 1986; Ruben et al., 1986). Despite the importance and prevalence of bursting neurons, however, in most cases little is known about the ionic mechanisms underlying their modulation. This is because few of these cells are well suited for voltage-clamp study.

The neurosecretory *Aplysia* cell R15 intrinsically generates robust bursting activity (Alving, 1968). Voltage-clamp studies have elucidated the ionic mechanism for producing bursting activity in this and other *Aplysia* neurons. It has been demonstrated that bursting activity is produced by the action of calcium current (I_{Ca}) and calcium-dependent currents (Gorman et al., 1982; Adams, 1985; Adams and Levitan, 1985; Kramer and Zucker, 1985a, b). Likewise, the actions of modulatory transmitters and cyclic nucleotides on R15 have been studied in detail (see Adams and Benson, 1985). For example, bath application of a membrane-permeable cyclic GMP analog (8-pcpt-cGMP) increases the number of action potentials per burst produced by cell R15 (Levitan and Norman, 1980).

Here, we report that 8-pcpt-cGMP can disrupt R15's bursting activity and cause the cell to produce tonic spiking or beating activity. Voltage-clamp studies reveal that 8-pcpt-cGMP reduces the steady-state subthreshold I_{Ca}. Voltage-clamp analysis further demonstrates that both the inward and outward calcium-dependent currents underlying bursting activity are also decreased by 8-pcpt-cGMP. It is suggested that the reduction of I_{Ca} and of calcium-dependent currents leads to a dampening of bursting that culminates in beating activity.

Materials and Methods

Materials and methods were similar to those reported in Levitan and Levitan (1988). The 8-pcpt-cGMP was synthesized as described by Miller et al. (1973), and was bath applied at a concentration of 1.7 mM for 10 min unless otherwise stated. Its effects were reversible upon washing. All experiments reported here were replicated at least 3 times, and the data shown are representative.

Simulation of bursting activity under voltage clamp consisted of presenting trains of 10-msec-long 100 mV depolarizing pulses, separated by 0.5–2 sec, alternated with long periods of inactivity at the holding potential of −50 mV. Typically, a burst cycle of 40 sec was used. Currents were typically filtered at 30 Hz.

Results

8-pcpt-cGMP dampens bursting activity produced by R15

Shown in Figure 1*A* is the basal bursting activity of cell R15. Upon bath application of 8-pcpt-cGMP, the first change in activity seen is an increase in the interval between action potentials within the burst (Fig. 1*B*). Later, the interburst hyperpolarizations become shallower (Fig. 1, *C, D*). Long bursts with many action potentials are produced, as reported previously (Levitan

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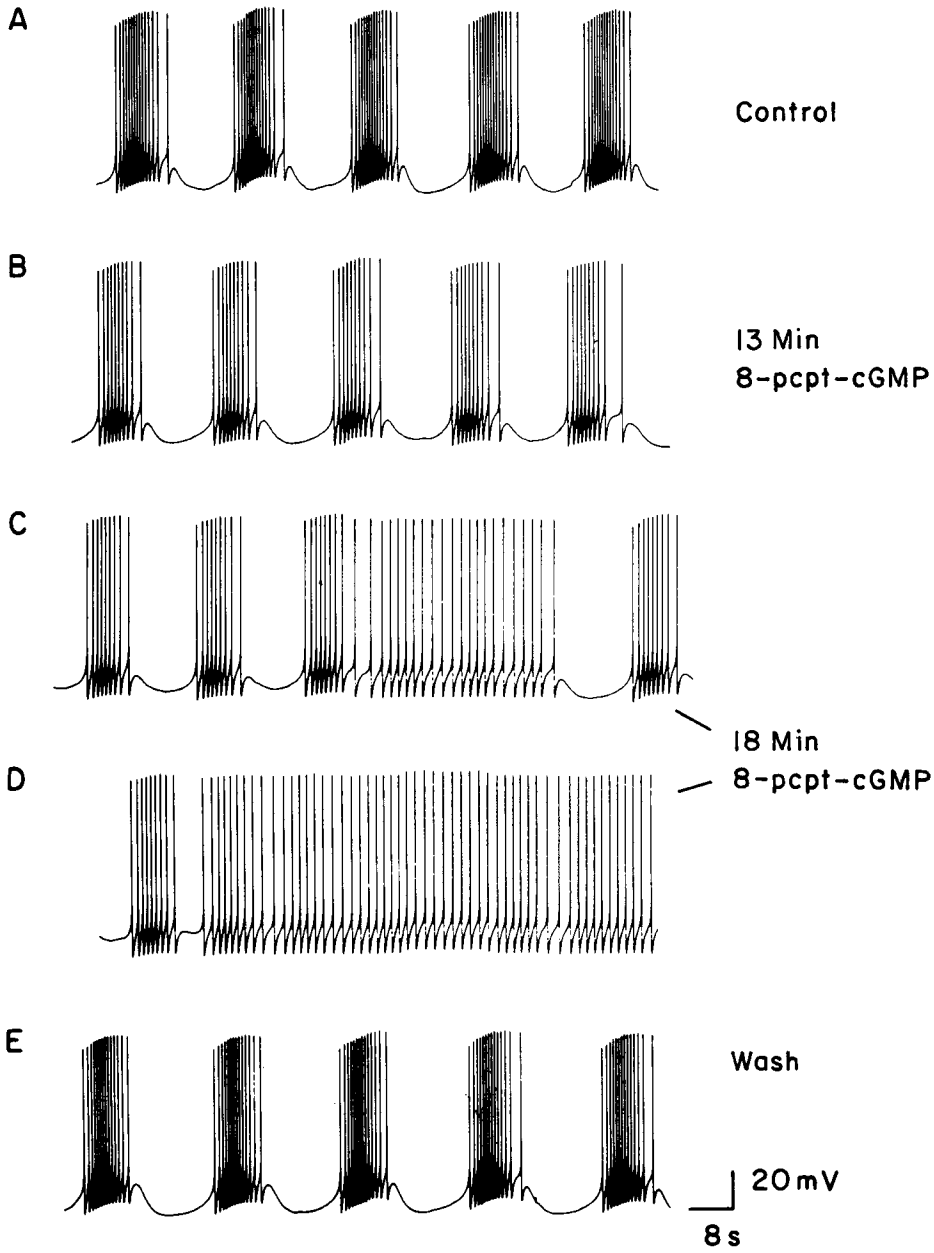


Figure 1. Effect of 0.85 mM 8-pcpt-cGMP on the bursting activity of R15. *A*, Control activity. *B*, After 13 min perfusion with 8-pcpt-cGMP, a decrease in spike frequency during the burst is seen. *C*, After 18 min, a depolarization of the interburst hyperpolarization is evident and long bursts become apparent. *D*, Continuous record from *C*. The final burst lasted many minutes. *E*, On washout the effects of 8-pcpt-cGMP reversed.

and Norman, 1980), but spiking occurs at a very low frequency (Fig. 1, *C, D*). Eventually, the cell stops bursting and instead produces slow tonic spiking or beating activity (Fig. 1*D*). The effects of 8-pcpt-cGMP are reversible upon washing (Fig. 1*E*). Thus, 8-pcpt-cGMP, like cyclic AMP under some conditions (Levitan and Levitan, 1988), can convert R15 from a burster to a beater. Unlike the effect of cyclic AMP, however, this is accompanied by a *decrease* in action potential frequency. Thus, it does not appear that 8-pcpt-cGMP acts simply by depolarizing R15.

The effects of 8-pcpt-cGMP take many minutes to develop and nearly an hour to reverse upon washing. They are not mimicked by bath application of equivalent concentrations of cyclic GMP or 8-bromo-cyclic GMP, both of which are less membrane-permeable than 8-pcpt-cGMP (data not shown). Furthermore, dibutyryl-cGMP, a compound known for *not* activating cyclic GMP-dependent protein kinase (Meyer and Miller, 1974) or cyclic GMP-gated ion channels in the vertebrate photoreceptor (Zimmerman et al., 1985), does not produce effects similar to 8-pcpt-cGMP, despite the fact that it is membrane permeable (data not shown). These data suggest that 8-pcpt-GMP acts intracellularly at a specific site to modulate bursting activity.

8-pcpt-cGMP decreases the voltage-gated calcium current

Voltage-clamp studies first addressed whether 8-pcpt-cGMP affects subthreshold currents that are known to play a part in burst production (Adams, 1985; Adams and Levitan, 1985; Kramer and Zucker, 1985a, b). Shown in Figure 2*A* is a current versus voltage (*I–V*) curve from a cell before and after exposure to 8-pcpt-cGMP. 8-pcpt-cGMP induces an outward voltage-gated current, which usually begins to activate between −75 and −60 mV. No reversal is seen, even at voltages very much more negative than the K^+ equilibrium potential (i.e., −75 mV). The

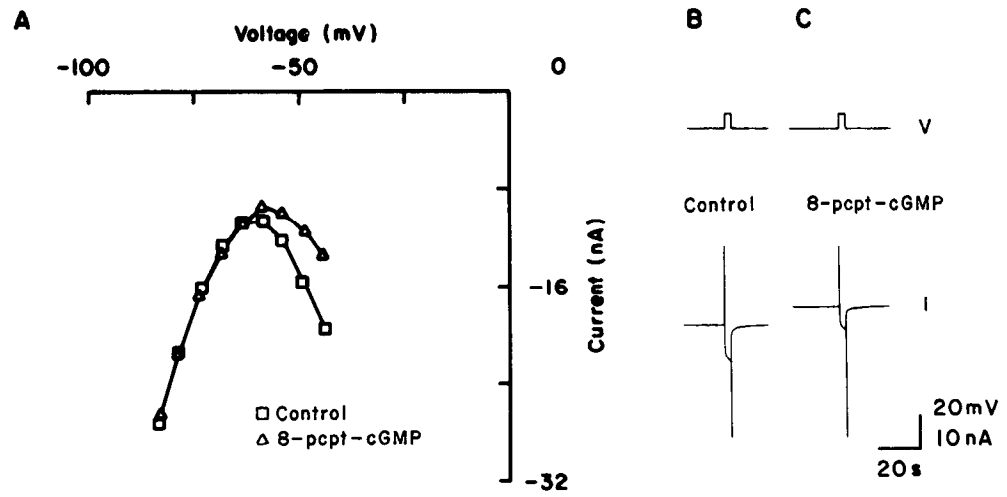


Figure 2. 8-pcpt-cGMP reduces the subthreshold inward current. *A*, Current versus voltage (*I-V*) curve from a cell before (□) and after (△) a 10 min perfusion with 1.7 mM 8-pcpt-cGMP. Currents were evoked by 0.5 sec pulses from −70 mV. *B*, Current (*lower trace*) evoked by depolarization (*upper trace*) from −52 mV in low-calcium SW. *C*, Same cell as *B* after a 10 min exposure to 1.7 mM 8-pcpt-cGMP.

8-pcpt-cGMP-induced current could be due to an increase in outward potassium or chloride currents, or to a decrease in the resting inward current, which is known to be carried by calcium (Gorman et al., 1982; Lewis et al., 1984).

This net outward current is also seen under conditions that dramatically reduce synaptic activity but leave the subthreshold I_{Ca} partially intact. Shown in Figure 2*B* is the current evoked by a small depolarizing voltage pulse in a cell bathed in a low Ca SW [see Levitan and Levitan (1988) for the ionic composition of this and other bathing solutions]. Applying 8-pcpt-cGMP shifts the holding current at −52 mV 5 nA outward and decreases the voltage-gated current by 43% (Fig. 2*C*). Thus, 8-pcpt-cGMP induces net outward current by acting directly on R15.

Analysis of the tail currents that follow depolarizing pulses favors the hypothesis that the cyclic GMP analog decreases resting I_{Ca}. If 8-pcpt-cGMP increased an outward potassium or chloride current, one would expect that the outward tail current following a depolarizing pulse would become larger in the presence of 8-pcpt-cGMP. In contrast, we have found that 8-pcpt-cGMP *decreases* the slow outward tail current evoked by long subthreshold depolarizing pulses (see arrows in Fig. 3, *A, B*). This outward tail current is due to inactivation of resting inward I_{Ca} by accumulation of internal calcium during the depolarizing pulse (Adams and Levitan, 1985; Kramer and Zucker, 1985b).

The extent of the decrease in the outward tail current is similar to the decrease in the voltage-gated inward current during the pulse. For example, in the representative cell shown in Figure 3, *A* and *B*, the voltage-gated inward current during the pulse is reduced by 42%, while the slow outward tail current following the pulse is reduced by 50%. 8-pcpt-cGMP also reduces the slow outward tail current that follows trains of short suprathreshold pulses (see below). These results support the hypothesis that 8-pcpt-cGMP decreases I_{Ca} and, therefore, decreases the calcium-dependent inactivation of resting I_{Ca} underlying the slow outward tail current.

To further test whether 8-pcpt-cGMP acts on I_{Ca}, calcium was substituted with manganese. Shown in Figure 3*A* is the basal voltage-gated subthreshold inward current in normal saline. Upon application of 8-pcpt-cGMP this current decreases (Fig. 3*B*). After washing out the 8-pcpt-cGMP, calcium was substituted with manganese. This blocks the subthreshold calcium current (Fig. 3*C*). Application of 8-pcpt-cGMP in the presence of manganese produces no outward current (Fig. 3*D*). Thus, blocking I_{Ca} with manganese also blocks the 8-pcpt-cGMP-induced voltage-gated outward current. This result further suggests that 8-pcpt-cGMP acts on I_{Ca} or on a calcium-dependent current.

To differentiate between these 2 possibilities, calcium-activated currents were blocked by loading R15 with the calcium chelator EGTA. Depolarization still evokes the voltage-gated I_{Ca}, although the calcium-dependent outward tail current is blocked (Fig. 3*E*). Application of 8-pcpt-cGMP continues to reduce the voltage-gated inward current (Fig. 3*F*). Thus, 8-pcpt-cGMP cannot be acting by increasing a calcium-activated outward current such as a calcium-activated potassium current. This also demonstrates that 8-pcpt-cGMP does not act by increasing calcium-dependent inactivation of the calcium current.

The data presented thus far do not eliminate the possibility that 8-pcpt-cGMP activates a manganese blockable voltage-gated potassium current. In fact, a potassium current that is not activated by intracellular calcium has been shown to be sensitive to divalent cations (Paupardin-Tritsch et al., 1981). To rule out activation of potassium channels by 8-pcpt-cGMP, the effects of 8-pcpt-cGMP were compared in normal (10 mM) and elevated (25 mM) potassium. Reducing the driving force for potassium approximately 2-fold does not alter the 8-pcpt-cGMP-induced outward current (Fig. 4). Thus, 8-pcpt-cGMP does not appear to be acting by increasing potassium current.

Dopamine has been shown to decrease axonal I_{Ca} in R15 (Lewis et al., 1984). If dopamine and 8-pcpt-cGMP act on the same ion channels, then dopamine should occlude the 8-pcpt-cGMP effect. Figure 5 shows the amplitude of the outward current induced by 8-pcpt-cGMP at different membrane potentials. These data were obtained by subtracting the *I-V* curve in the presence of 8-pcpt-cGMP from the *I-V* curve after washing away the 8-pcpt-cGMP. The cell was then again exposed to 8-pcpt-cGMP but this time in the presence of dopamine, and again a difference curve is shown. This difference current is due in part to the desensitization of the effects of saturating dopamine during the long time course of the experiment. The recovery of dopamine-blocked I_{Ca} (due to desensitization) produces an outward difference current that sums with the 8-pcpt-cGMP-induced outward current. Despite this overestimation of the 8-pcpt-cGMP-induced current, the net outward current produced by 8-pcpt-cGMP is much smaller in the presence of do-

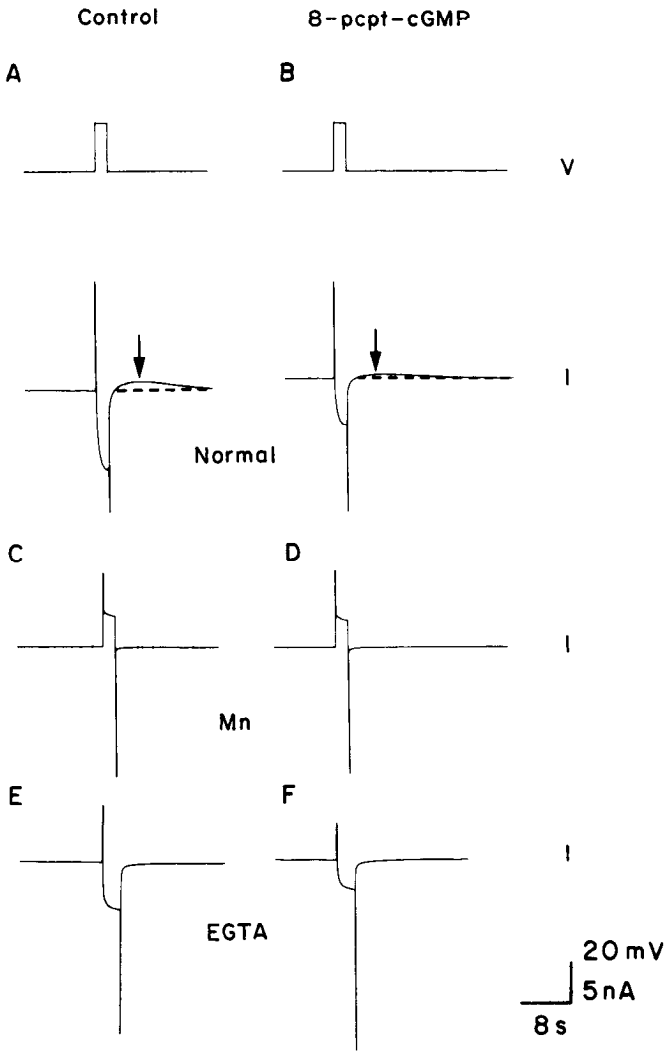


Figure 3. Effects of manganese and internal EGTA on the 8-pcpt-cGMP-induced reductions in the voltage-gated inward current and the outward tail current. *A*, Current (*bottom*) evoked by a 24 mV depolarizing pulse (*top*) from −70 mV. *Arrow* indicates slow tail current, which is outward compared with the holding current (*dotted line*). *B*, 8-pcpt-cGMP reduces the voltage-gated inward current and the slow outward tail current. *C*, After washout of the 8-pcpt-cGMP, manganese was substituted for calcium. Depolarization now evokes a net outward current. *D*, 8-pcpt-cGMP produces no change in the presence of manganese. *E*, Current evoked by a 22 mV depolarization from −70 mV in an EGTA-loaded cell. *F*, 8-pcpt-cGMP reduces the voltage-gated inward current even in the presence of internal EGTA.

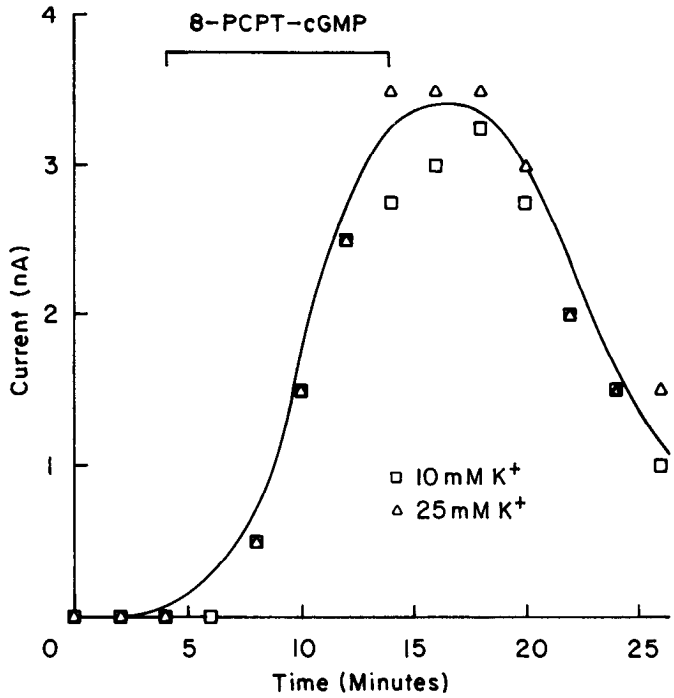


Figure 4. Effect of 8-pcpt-cGMP in normal and elevated potassium. The time during which 8-pcpt-cGMP was applied is indicated by the *bracket*. The net outward current induced by 8-pcpt-cGMP at −29 mV in 10 mM (normal) potassium (□) is not affected by elevating potassium to 25 mM (△). Similar results were also obtained in cells where effects of 8-pcpt-cGMP were measured at −40 mV. Low-calcium seawater was used to prevent synaptic activation by the elevated potassium. The curve was drawn by eye.

pamine than in its absence. Normally, 8-pcpt-cGMP gives similar responses with multiple applications (data not shown). Thus, dopamine appears to at least partially occlude the effects of 8-pcpt-cGMP. These data further support the hypothesis that 8-pcpt-cGMP decreases I_{Ca} in R15.

We cannot completely eliminate the possibility that 8-pcpt-cGMP increases chloride currents. This is because the long time course of application and removal of 8-pcpt-cGMP (i.e., 60 min) allows chloride to redistribute during an experiment in which the chloride equilibrium potential is altered by lowering external chloride. However, it seems highly unlikely that 8-pcpt-cGMP acts on a chloride current. First, increasing chloride current should increase the slow outward tail current, whereas 8-pcpt-cGMP decreases the slow outward tail current. Furthermore, we have found that 8-pcpt-cGMP induces outward current at potentials as low as −72 mV, while the chloride equilibrium potential in *Aplysia* neurons is approximately −58 mV (Russell and Brown, 1972; Ascher et al., 1976). Finally, no chloride channels have been reported to be blocked by manganese. Thus, we conclude that 8-pcpt-cGMP decreases I_{Ca}.

Analysis of 8-pcpt-cGMP effects in the context of bursting

The finding that 8-pcpt-cGMP decreases I_{Ca} presents a quandry. Outward currents induced by transmitters or injected under current clamp lead to enhancement of the interburst hyperpolarization (Frazier et al., 1967; Wilson and Wachtel, 1978; Ayers and Selverston, 1979; Benson and Levitan, 1983). How can induction of tonic spiking be associated with a decrease in inward I_{Ca}? A decrease in inward current is, after all, equivalent to increasing outward current. The voltage-clamp analysis presented thus far employs simple voltage paradigms where the effects of 8-pcpt-cGMP are analyzed under steady-state conditions. However, R15 usually is not at steady state. Rather, the cell produces oscillatory activity, and oscillatory systems are far from equilibrium and steady-state conditions. Thus, in order to gain an understanding of how 8-pcpt-cGMP alters bursting, a voltage-clamp paradigm was used in which trains of short spike-like depolarizations were alternated with 40 sec periods at the holding potential. In effect, the cell was commanded to burst but at a spike frequency and interburst period set by the experimenter. While this paradigm does not exactly mimic endogenously generated bursting, it approximates it while still allowing for analysis of the effects of 8-pcpt-cGMP under voltage clamp.

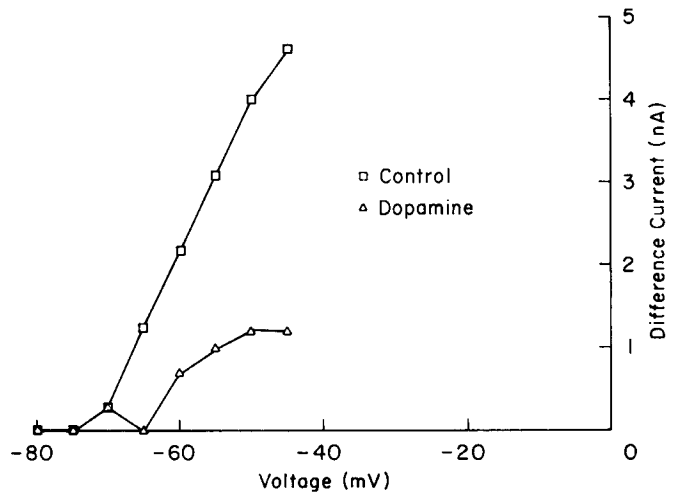


Figure 5. Dopamine reduces the 8-pcpt-cGMP-induced voltage-gated current. Shown are the 8-pcpt-cGMP-induced currents in the absence (□) and presence (△) of 0.5 mM dopamine. Difference curves were obtained by subtracting the *I-V* curve in the presence of 8-pcpt-cGMP from the *I-V* curve after washing out 8-pcpt-cGMP.

Action potentials and depolarizing pulses that cause calcium influx have been shown to produce 2 slow currents important for generating oscillatory activity in molluscan bursting cells. First, a calcium-activated inward current is seen that peaks approximately 500 msec after an action potential (see upward arrows in Fig. 6). This current, which is thought to be due to the opening of nonselective cation channels in the axon of R15, contributes to maintaining the burst and underlies the depolarizing afterpotential seen after the last spike in the burst (Kramer and Zucker, 1985a; Adams and Levitan, 1985; Kramer, 1986). Second, at the end of a "burst" of depolarizing pulses, an outward current with slow kinetics is seen (see downward arrows in Fig. 6). This current, which underlies the interburst hyperpolarization, is due to calcium-dependent inactivation of subthreshold calcium current (Adams and Levitan, 1985; Kramer and Zucker, 1985b). It has been shown that, because of the large difference in the kinetics of these currents, the calcium-activated cation current after the first pulse in a train is relatively uncontaminated by the slow inactivation current. Likewise, the slow inactivation current seen 2–4 sec after the end of a burst is uncontaminated by the nonselective cation current, which decays more quickly (Adams, 1985).

The effects of 8-pcpt-cGMP on these voltage-clamp-induced "burst" currents in a representative cell are shown in Figure 6*B*. 8-pcpt-cGMP induces a number of changes. First, the holding current evident before each burst becomes more outward. This is consistent with the finding that 8-pcpt-cGMP decreases steady-state I_{Ca}. Additionally, a 44% decrease in the pulse-evoked inward current is seen. This may simply be due to a decrease in I_{Ca}, which, in turn, leads to a reduction in the calcium-activated nonselective cation current. On the other hand, 8-pcpt-cGMP might increase some opposing outward current or affect the cation channels directly. The predominant outward current active 500 msec after an action potential is the calcium-activated potassium current. This current is blocked by low concentrations of tetraethylammonium (TEA; K_d = 0.4 mM) (Hermann and Gorman, 1981). Figure 6, *C* and *D*, shows that 8-pcpt-cGMP reduces the spike-evoked inward current in the presence of concentrations of TEA that block greater than 92% of the calcium-activated potassium current, and thus 8-pcpt-cGMP does not act by increasing that current. The fact that the calcium-activated nonselective cation current is located predominantly in the axon of R15 prevented further study on its modulation by 8-pcpt-cGMP. Thus, we cannot determine whether 8-pcpt-cGMP decreases this current directly or indirectly, or whether some opposing current other than the calcium-activated potassium current is modulated. In either case, we are still left with the apparent paradox that conversion of bursting to beating activity by 8-pcpt-cGMP is accompanied by a decrease in inward currents. The reduction of I_{Ca} and spike-evoked calcium-dependent inward current tends to remove the drive to spike vigorously and thus probably causes the decrease in spike frequency seen with 8-pcpt-cGMP.

In order to analyze fully the effects of 8-pcpt-cGMP one must also consider the effects of 8-pcpt-cGMP on the current terminating a burst. Normally, calcium-dependent inactivation of the subthreshold calcium current leads to the cessation of a burst (Adams and Levitan, 1985; Kramer and Zucker, 1985b). In fact, 8-pcpt-cGMP reduces by some 50% the slow outward tail current which reflects the inactivation of the calcium current. This is evident in experiments in which the slow outward tail current is evoked either by a long subthreshold depolarization or by trains of short spikelike depolarizations (see downward arrows in Figs. 3 and 6). This decrease in the current responsible for producing the interburst hyperpolarization leads to the shallow interbursts seen with 8-pcpt-cGMP.

Thus far, the currents underlying bursting have been analyzed in a context in which spike frequency is fixed. However, 8-pcpt-cGMP decreases the inward currents that control spike frequency. We therefore tested whether varying spike frequency alters the currents that produce bursting. Shown in Figure 7*A* are the currents evoked by a train of pulses given with an interpulse interval of 0.5 sec in a representative cell. Large inward, as well as outward, currents are seen. Doubling the interpulse interval does not affect the inward current seen after the first pulse (the only one not contaminated by the slow outward current) and only reduces the peak inward current, seen after the third pulse, by 18%. However, the slow outward tail current seen at the end of the train of pulses is reduced by 45% (Fig. 7*B*). A further decrease in the outward tail current is seen when the interpulse interval is increased to 1.5 sec, but once again the inward tail current after the first pulse is unchanged (Fig. 7*C*). These changes are evident in the plot shown in Figure 7*D*. There we see that increasing the interpulse interval, from a period commonly seen in control cells to a period seen in cells exposed to 8-pcpt-cGMP, markedly reduces the outward current responsible for terminating a burst. This finding is not surprising since increasing the interval between spikes decreases temporal summation of the outward current. Thus, combining the reduction in the slow outward current produced directly by 8-pcpt-cGMP (Fig. 6), with the indirect reduction caused by decreasing spike frequency (Fig. 7), leads to a profound decrease in this outward current responsible for terminating the burst.

These results suggest that 8-pcpt-cGMP may act by the following mechanism to convert R15's activity from robust bursting to slow beating. 8-pcpt-cGMP decreases the subthreshold I_{Ca} and the action potential-induced inward and outward currents. The reduction of inward currents, in turn, causes a decrease in spike frequency that further diminishes the outward current responsible for terminating the burst. Reducing this out-

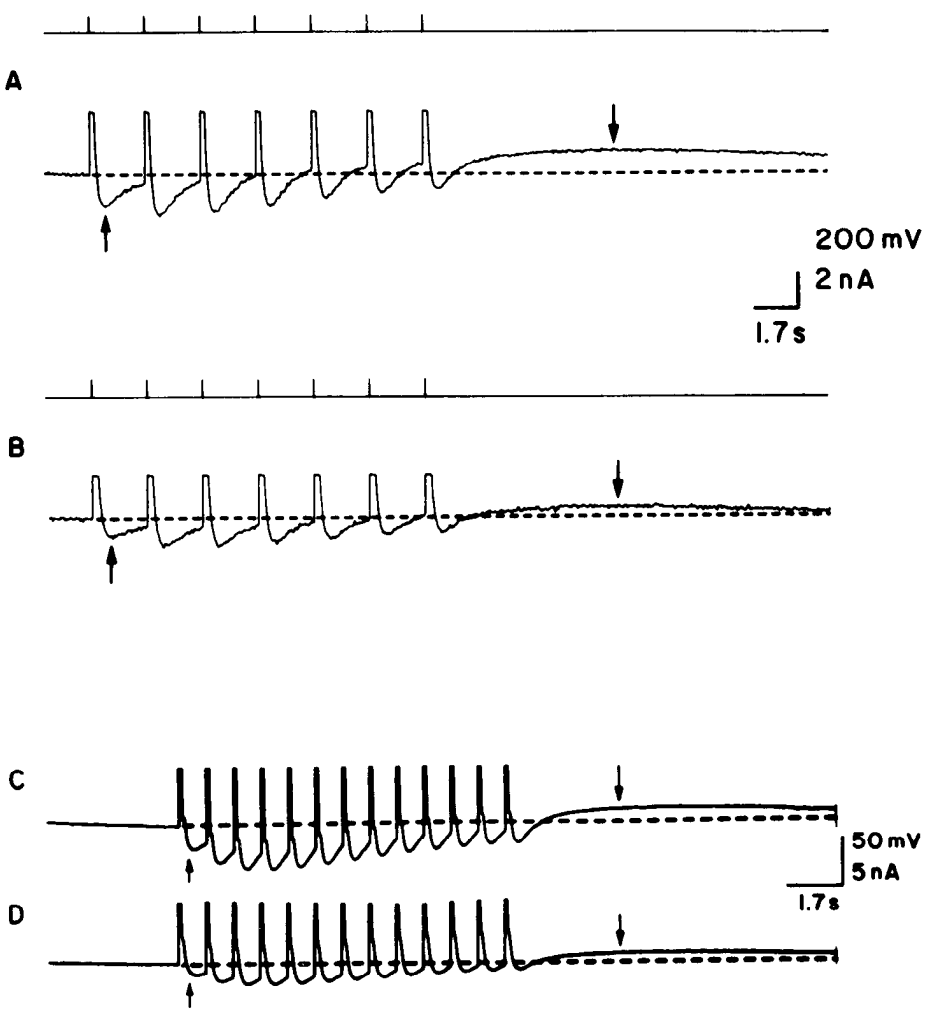


Figure 6. 8-pcpt-cGMP decreases the burst-evoked inward and outward currents. *A*, Trains of 10 msec depolarizations (*upper traces* in *A* and *B*) from −50 mV to +50 mV evoke inward tail currents (*upward arrow*), followed by a slow outward tail current (*downward arrow*) at the end of the simulated "burst" of depolarizations. *B*, Perfusion for 18 min with 0.85 mM 8-pcpt-cGMP decreases the inward (*upward arrow*) and outward (*downward arrow*) tail currents. The holding current before the "burst" shifted 2.5 nA outward in the presence of 8-pcpt-cGMP. *C*, Burst-evoked currents in another cell bathed in 5 mM TEA. *D*, A 10 min perfusion with 1.7 mM 8-pcpt-cGMP reduces both the inward and outward tail currents in the continued presence of TEA. The current before the "burst" shifted 4 nA outward.

ward current decreases the interburst hyperpolarization. Eventually, the outward current becomes so small that it is insufficient to produce a long-lasting interburst hyperpolarization. The cell is left with steady-state voltage-gated and spike-evoked inward currents, which, although reduced, are still sufficient to maintain beating activity. If this scheme is correct, 8-pcpt-cGMP does not act simply to depolarize or hyperpolarize R15; rather, 8-pcpt-cGMP converts bursting to beating activity by reducing both the inward and outward currents required for bursting.

8-pcpt-cGMP alters R15's responses to inhibitory and excitatory input

R15 is bombarded by a wide variety of synaptic potentials when the nervous system is left intact (see Adams and Benson, 1985). In order to determine the physiological significance of 8-pcpt-cGMP-induced beating activity, one must determine whether 8-pcpt-cGMP changes R15's responsiveness to synaptic input. As a first step in this direction, the effects of injected current pulses were compared in the absence and presence of 8-pcpt-cGMP. While these stimuli are not physiological, applying current directly to R15 avoids the need to correct for presynaptic effects. Presumably, some synaptic currents will produce effects similar to those seen with current pulses.

Shown in Figure 8 are the responses of R15 to hyperpolarizing and depolarizing current pulses. Very little change in the basal bursting behavior is seen with either small depolarizing or hyperpolarizing current pulses, delivered either during the burst or during the interburst hyperpolarization (Fig. 8, *A, B*). 8-pcpt-cGMP was then applied to the cell, and the effects of identical pulses were determined. A hyperpolarizing current pulse now initiates a long-lasting hyperpolarization (Fig. 8*C*), presumably because the hyperpolarizing pulse turns off the small amount of voltage-gated I_{Ca} available to depolarize the cell. Dramatic hyperpolarization is also seen with *depolarizing* pulses which increase spike frequency, allow calcium entry, and thus increase the calcium-dependent inactivation of the small amount of resting I_{Ca} (Fig. 8, *D, E*). Current pulses also can sometimes induce dampened bursting activity (Fig. 8*F*). These results demonstrate that the 8-pcpt-cGMP-induced reduction of subthreshold currents that underlie bursting activity makes R15 very sensitive to perturbation. A physiological stimulus that produces 8-pcpt-cGMP-like effects would therefore not only alter basal bursting activity, but would also effectively sensitize R15 to synaptic input.

Discussion

We have found that 8-pcpt-cGMP can convert cell R15's bursting activity into slow beating activity. Voltage-clamp studies reveal that 8-pcpt-cGMP reduces the steady-state subthreshold I_{Ca}. Furthermore, voltage-clamp paradigms that mimic bursting activity demonstrate that the spike-evoked inward and outward currents that underlie bursting are also decreased. The decreases

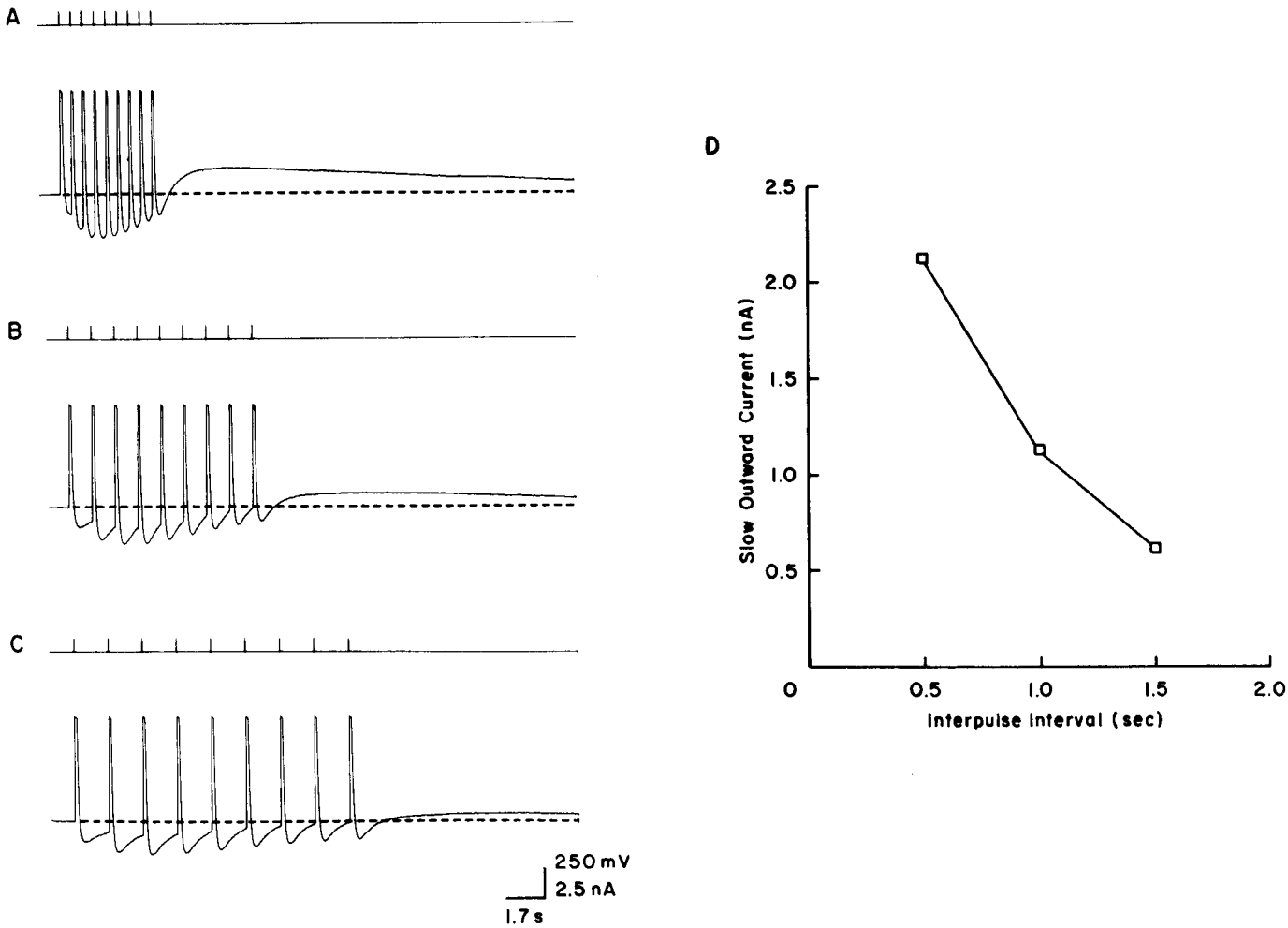


Figure 7. Effect of varying the interpulse interval on the slow outward tail current following a simulated burst produced as described for Figure 6. *Top trace* in each pair is the train of voltage pulses; *bottom trace,* evoked current. *A,* Currents evoked by a train of pulses with an interpulse interval of 0.5 sec. *B,* Currents evoked with interpulse interval of 1.0 sec. *C,* Currents evoked with interpulse interval of 1.5 sec. *D,* Effect of interpulse interval on the peak slow outward tail current. Note that increasing the interpulse interval dramatically reduces the slow outward tail current.

in these 2 currents may be due merely to the fact that they are calcium dependent and thus respond indirectly to the reduction of I_{Ca}. These findings suggest that 8-pcpt-cGMP does not simply hyperpolarize or depolarize R15 but rather dampens bursting. The cell beats slowly because inward currents are too small to drive high-frequency spiking, and the outward current is too small to produce an interburst hyperpolarization.

To test this hypothesis we employed a computer model originally formulated to demonstrate that the spike-evoked currents, when combined with a steady-state inward current, are sufficient to produce bursting activity (Adams, 1985). When physiological parameters are given for the resting steady-state current and the spike-evoked tail currents, a bursting pattern reminiscent of that in R15 is produced (results not shown). Reducing both the inward and outward currents by 25% leads to production of bursts with lower spike frequencies and shallow interbursts similar to those seen in Figure 1*C*. Further reduction of the burst currents to 50% of their initial values gives tonic low-frequency beating activity. Thus, decreasing the opposing inward and outward burst currents to a degree seen with 8-pcpt-cGMP leads the model to produce the same activity seen with the cyclic GMP analog. This suggests that the hypothesis that bursting activity can be converted to beating activity by reducing both the inward and outward currents required for production of oscillatory activity is a viable one.

Further support for this hypothesis is provided by examining the response of R15 to stimulation of the branchial nerve. This elicits an inhibitory synapse (Parnas et al., 1974) that appears to reduce the same currents in R15 as does 8-pcpt-cGMP, albeit to a larger extent (Adams et al., 1980; Adams and Benson, 1985; R. Kramer, personal communication). Recovery from the synaptic hyperpolarization is associated with dampened bursting, and this pattern of activity displays many of the features seen in the presence of 8-pcpt-cGMP (Parnas et al., 1974; E. S. Levitan and I. B. Levitan, unpublished observations). This result suggests that conversion of a burster to a slow beater can occur with physiological stimuli that reduce opposing burst currents.

The dampening of bursting produced by 8-pcpt-cGMP also amplifies R15's responsiveness to hyperpolarizing currents that are either injected experimentally or evoked by increasing spike frequency. This change most likely arises because the cyclic GMP analog decreases the inward currents that normally oppose a hyperpolarization and bring the cell back to the spike threshold. Synaptic potentials may likewise produce larger alterations

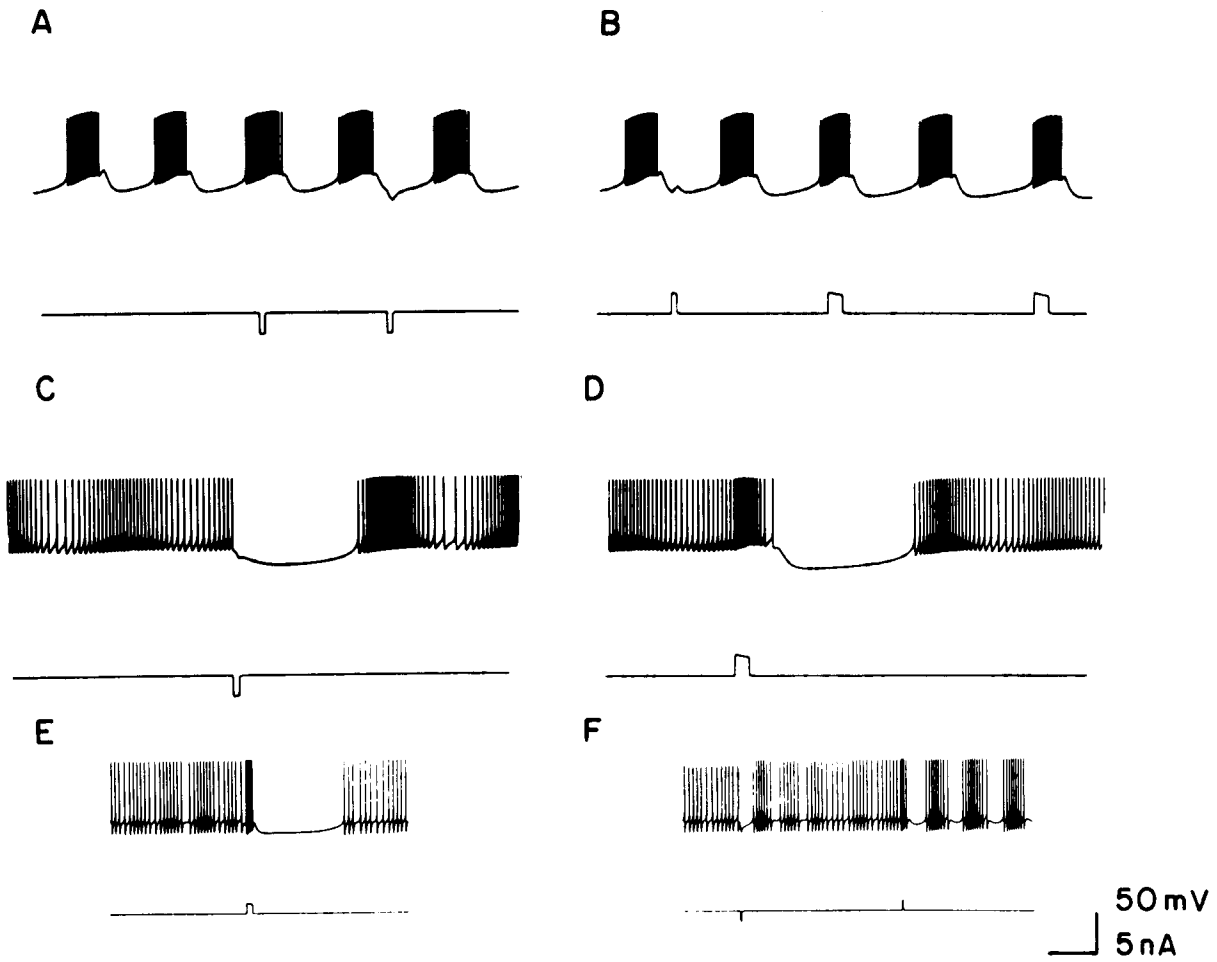


Figure 8. 8-pcpt-cGMP alters the responsiveness of R15 to inhibitory and excitatory currents. *A*, Hyperpolarizing current pulses (*lower trace*), delivered during either the burst or interburst, produce little effect on R15's bursting activity (*upper trace*). *B*, Similarly, depolarizing current pulses do not alter the basal bursting activity. After induction of beating activity by 8-pcpt-cGMP, a hyperpolarizing current pulse (*C*) or a depolarizing pulse (*D*) can both induce a long-lasting hyperpolarization. *E*, Same as *D* in another cell (tops of spikes clipped off). *F*, Same cell as *E*. Shorter hyperpolarizing or depolarizing pulses can trigger transient periods of bursting. Eventually, resumption of beating activity is seen. Time calibration: 8 sec, *A–D*; 20 sec, *E* and *F*.

in R15's activity in the presence of 8-pcpt-cGMP-like stimuli. This is reminiscent of the effects of 5-HT, which acts via cyclic AMP to alter R15's responsiveness to current pulses (Levitan and Levitan, 1988). While 8-pcpt-cGMP and cyclic AMP act differently, both modulate opposing currents and produce effects that cannot be termed simply excitatory or inhibitory. Both serve also to illustrate that modulatory mechanisms, which produce straightforward changes in activity in normally silent cells that are near equilibrium as far as the membrane potential is concerned, can produce complicated behavior in an oscillatory system such as a neuronal burster.

It is interesting that cyclic AMP and 8-pcpt-cGMP produce apparently opposite effects on the subthreshold I_{Ca}. We cannot be certain that the effects of 8-pcpt-cGMP represent the physiological action of cyclic GMP. However, 8-position-substituted cyclic GMP analogs have been shown to mimic some actions of cyclic GMP (Miller et al., 1973; Zimmerman et al., 1985). Thus, it seems possible that cyclic AMP increases and cyclic GMP decreases I_{Ca} in R15. A similar situation occurs in heart, where cyclic GMP activates a phosphodiesterase that, in turn, lowers cyclic AMP and thus decreases I_{Ca} (Hartzell and Fischmeister, 1986). However, it is unlikely that 8-pcpt-cGMP acts by stimulating a phosphodiesterase in R15. First, 8-position-substituted cyclic GMP analogs are poor activators of the cyclic GMP-dependent phosphodiesterase (Erneux et al., 1985). More important, no change in I_R is evoked by 8-pcpt-cGMP. This current, like I_{Ca}, is modulated by cyclic AMP (Drummond et al., 1980; Benson and Levitan, 1983; Lemos and Levitan, 1984), and thus a change in cyclic AMP concentration would be expected to affect it also. Of course, this argument assumes that cyclic GMP-activated phosphodiesterase, calcium channels, and inwardly rectifying potassium channels are evenly distributed throughout the cell.

8-pcpt-cGMP might also act by activating cyclic GMP-dependent protein kinase, which occurs in the *Aplysia* nervous system (Novak-Hofer et al., 1985). On the other hand, it might modulate channel gating in an ATP-independent manner, as it does in vertebrate rods (Haynes et al., 1986; Zimmerman and Baylor, 1986). It is unlikely that 8-pcpt-cGMP alters bursting by cross-reacting with cyclic AMP-dependent protein kinase or by inhibiting a phosphodiesterase since the effects of 8-pcpt-cGMP differ from those of cyclic AMP. In any case, the recent finding that cyclic GMP acts via cyclic GMP-dependent protein kinase to *increase* I_{Ca} in certain *Helix* neurons (Paupardin-Tritsch

et al., 1986a, b) suggests that cyclic GMP, like cyclic AMP, produces varying effects in different cell types. This is reinforced by the fact that cyclic GMP increases a sodium current in some molluscan neurons (Connor and Hockberger, 1984).

The effects of 5-HT and 8-pcpt-cGMP on R15 suggest that a burster can be converted into a beater by at least 2 mechanisms. 5-HT, 50 μM, increases the subthreshold I_{Ca} so that the cell always returns to the action potential threshold (Levitan and Levitan, 1988), whereas 8-pcpt-cGMP dampens oscillatory activity until the cell is left beating slowly. These changes in activity may also alter the cell's responsiveness to synaptic input. While we have not determined whether cyclic GMP is a physiological second messenger in R15, it is interesting that the conversion of cell 11 in *Otala* from beating to bursting activity by vasopressin is associated with an increase in steady-state inward current. Injection of inward current, however, does not produce bursting (Barker and Smith, 1976). Perhaps, in analogy to recovery of bursting seen when washing away 8-pcpt-cGMP from R15, bursting is induced in cell 11 by increasing the steady-state inward current and the action potential-evoked outward current. This mechanism may also be important for interconversion between bursting and beating activity in cells that are not well suited for voltage-clamp analysis (Grace and Bunney, 1984; Marder and Eisen, 1984; Dekin et al., 1985). For example, in nigral dopamine neurons, exposure to the excitatory transmitter glutamate shifts the cells from a beating to a bursting state (Grace and Bunney, 1984). It will be of interest to extend the present analysis to some of these other systems to determine the generality of these ionic mechanisms.

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